

13. Prove that a group of order 56 has a normal Sylow p -subgroup for some prime p dividing its order.
14. Prove that a group of order 312 has a normal Sylow p -subgroup for some prime p dividing its order.
15. Prove that a group of order 351 has a normal Sylow p -subgroup for some prime p dividing its order.
16. Let $|G| = pqr$, where p, q and r are primes with $p < q < r$. Prove that G has a normal Sylow subgroup for either p, q or r .
17. Prove that if $|G| = 105$ then G has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.
18. Prove that a group of order 200 has a normal Sylow 5-subgroup.
19. Prove that if $|G| = 6545$ then G is not simple.
20. Prove that if $|G| = 1365$ then G is not simple.
21. Prove that if $|G| = 2907$ then G is not simple.
22. Prove that if $|G| = 132$ then G is not simple.
23. Prove that if $|G| = 462$ then G is not simple.
24. Prove that if G is a group of order 231 then $Z(G)$ contains a Sylow 11-subgroup of G and a Sylow 7-subgroup is normal in G .
25. Prove that if G is a group of order 385 then $Z(G)$ contains a Sylow 7-subgroup of G and a Sylow 11-subgroup is normal in G .
26. Let G be a group of order 105. Prove that if a Sylow 3-subgroup of G is normal then G is abelian.
27. Let G be a group of order 315 which has a normal Sylow 3-subgroup. Prove that $Z(G)$ contains a Sylow 3-subgroup of G and deduce that G is abelian.
28. Let G be a group of order 1575. Prove that if a Sylow 3-subgroup of G is normal then a Sylow 5-subgroup and a Sylow 7-subgroup are normal. In this situation prove that G is abelian.
29. If G is a non-abelian simple group of order < 100 , prove that $G \cong A_5$. [Eliminate all orders but 60.]
30. How many elements of order 7 must there be in a simple group of order 168?
31. For $p = 2, 3$ and 5 find $n_p(A_5)$ and $n_p(S_5)$. [Note that $A_4 \leq A_5$.]
32. Let P be a Sylow p -subgroup of H and let H be a subgroup of K . If $P \trianglelefteq H$ and $H \trianglelefteq K$, prove that P is normal in K . Deduce that if $P \in \text{Syl}_p(G)$ and $H = N_G(P)$, then $N_G(H) = H$ (in words: *normalizers of Sylow p -subgroups are self-normalizing*).
33. Let P be a normal Sylow p -subgroup of G and let H be any subgroup of G . Prove that $P \cap H$ is the unique Sylow p -subgroup of H .
34. Let $P \in \text{Syl}_p(G)$ and assume $N \trianglelefteq G$. Use the conjugacy part of Sylow's Theorem to prove that $P \cap N$ is a Sylow p -subgroup of N . Deduce that PN/N is a Sylow p -subgroup of G/N (note that this may also be done by the Second Isomorphism Theorem — cf. Exercise 9, Section 3.3).
35. Let $P \in \text{Syl}_p(G)$ and let $H \leq G$. Prove that $gPg^{-1} \cap H$ is a Sylow p -subgroup of H for some $g \in G$. Give an explicit example showing that $hPh^{-1} \cap H$ is not necessarily a Sylow p -subgroup of H for any $h \in H$ (in particular, we cannot always take $g = 1$ in the first part of this problem, as we could when H was normal in G).

36. Prove that if N is a normal subgroup of G then $n_p(G/N) \leq n_p(G)$.
37. Let R be a normal p -subgroup of G (not necessarily a Sylow subgroup).
 (a) Prove that R is contained in every Sylow p -subgroup of G .
 (b) If S is another normal p -subgroup of G , prove that RS is also a normal p -subgroup of G .
 (c) The subgroup $O_p(G)$ is defined to be the group generated by all normal p -subgroups of G . Prove that $O_p(G)$ is the unique largest normal p -subgroup of G and $O_p(G)$ equals the intersection of all Sylow p -subgroups of G .
 (d) Let $\overline{G} = G/O_p(G)$. Prove that $O_p(\overline{G}) = \overline{1}$ (i.e., $\overline{G}$ has no nontrivial normal p -subgroup).
38. Use the method of proof in Sylow's Theorem to show that if n_p is not congruent to 1 (mod p^2) then there are distinct Sylow p -subgroups P and Q of G such that $|P : P \cap Q| = |Q : P \cap Q| = p$.
39. Show that the subgroup of strictly upper triangular matrices in $GL_n(\mathbb{F}_p)$ (cf. Exercise 17, Section 2.1) is a Sylow p -subgroup of this finite group. [Use the order formula in Section 1.4 to find the order of a Sylow p -subgroup of $GL_n(\mathbb{F}_p)$.]
40. Prove that the number of Sylow p -subgroups of $GL_2(\mathbb{F}_p)$ is $p + 1$. [Exhibit two distinct Sylow p -subgroups.]
41. Prove that $SL_2(\mathbb{F}_4) \cong A_5$ (cf. Exercise 9, Section 2.1 for the definition of $SL_2(\mathbb{F}_4)$).
42. Prove that the group of rigid motions in $\mathbb{R}^3$ of an icosahedron is isomorphic to A_5 . [Recall that the order of this group is 60: Exercise 13, Section 1.2.]
43. Prove that the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron is isomorphic to A_5 . (As with the cube and the tetrahedron, the icosahedron and the dodecahedron are dual solids.) [Recall that the order of this group is 60: Exercise 12, Section 1.2.]
44. Let p be the smallest prime dividing the order of the finite group G . If $P \in \text{Syl}_p(G)$ and P is cyclic prove that $N_G(P) = C_G(P)$.
45. Find generators for a Sylow p -subgroup of S_{2p} , where p is an odd prime. Show that this is an abelian group of order p^2 .
46. Find generators for a Sylow p -subgroup of S_{p^2} , where p is a prime. Show that this is a non-abelian group of order p^{p+1} .
47. Write and execute a computer program which
 (i) gives each odd number $n < 10,000$ that is not a power of a prime and that has some prime divisor p such that n_p is not forced to be 1 for all groups of order n by the congruence condition of Sylow's Theorem, and
 (ii) gives for each n in (i) the factorization of n into prime powers and gives the list of all permissible values of n_p for all primes p dividing n (i.e., those values not ruled out by Part 3 of Sylow's Theorem).
48. Carry out the same process as in the preceding exercise for all even numbers less than 1000. Explain the relative lengths of the lists versus the number of integers tested.
49. Prove that if $|G| = 2^n m$ where m is odd and G has a cyclic Sylow 2-subgroup then G has a normal subgroup of order m . [Use induction and Exercises 11 and 12 in Section 2.]
50. Prove that if U and W are normal subsets of a Sylow p -subgroup P of G then U is conjugate to W in G if and only if U is conjugate to W in $N_G(P)$. Deduce that two elements in the center of P are conjugate in G if and only if they are conjugate in $N_G(P)$. (A subset U of P is normal in P if $N_P(U) = P$.)

- 51.** Let P be a Sylow p -subgroup of G and let M be any subgroup of G which contains $N_G(P)$. Prove that $|G : M| \equiv 1 \pmod{p}$.

The following sequence of exercises leads to the classification of all numbers n with the property that every group of order n is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

- 52.** Suppose G is a finite simple group in which every proper subgroup is abelian. If M and N are distinct maximal subgroups of G prove $M \cap N = 1$. [See Exercise 23 in Section 3.]
- 53.** Use the preceding exercise to prove that if G is any non-abelian group in which every proper subgroup is abelian then G is not simple. [Let G be a counterexample to this assertion and use Exercise 24 in Section 3 to show that G has more than one conjugacy class of maximal subgroups. Use the method of Exercise 23 in Section 3 to count the elements which lie in all conjugates of M and N , where M and N are nonconjugate maximal subgroups of G ; show that this gives more than $|G|$ elements.]
- 54.** Prove the following classification: if G is a finite group of order $p_1 p_2 \dots p_r$ where the p_i 's are distinct primes such that p_i does not divide $p_j - 1$ for all i and j , then G is cyclic. [By induction, every proper subgroup of G is cyclic, so G is not simple by the preceding exercise. If N is a nontrivial proper normal subgroup, N is cyclic and G/N acts as automorphisms of N . Use Proposition 16 to show that $N \leq Z(G)$ and use induction to show $G/Z(G)$ is cyclic, hence G is abelian by Exercise 36 of Section 3.1.]
- 55.** Prove the converse to the preceding exercise: if $n \geq 2$ is an integer such that every group of order n is cyclic, then $n = p_1 p_2 \dots p_r$ is a product of distinct primes and p_i does not divide $p_j - 1$ for all i, j . [If n is not of this form, construct noncyclic groups of order n using direct products of noncyclic groups of order p^2 and pq , where $p \mid q - 1$.]
- 56.** If G is a finite group in which every proper subgroup is abelian, show that G is solvable.

4.6 THE SIMPLICITY OF A_n

There are a number of proofs of the simplicity of A_n , $n \geq 5$. The most elementary involves showing A_n is generated by 3-cycles. Then one shows that a normal subgroup must contain one 3-cycle hence must contain all the 3-cycles so cannot be a proper subgroup. We include a less computational approach.

Note that A_3 is an abelian simple group and that A_4 is not simple ($n_2(A_4) = 1$).

Theorem 24. A_n is simple for all $n \geq 5$.

Proof: By induction on n . The result has already been established for $n = 5$, so assume $n \geq 6$ and let $G = A_n$. Assume there exists $H \trianglelefteq G$ with $H \neq 1$ or G .

For each $i \in \{1, 2, \dots, n\}$ let G_i be the stabilizer of i in the natural action of G on $i \in \{1, 2, \dots, n\}$. Thus $G_i \leq G$ and $G_i \cong A_{n-1}$. By induction, G_i is simple for $1 \leq i \leq n$.

Suppose first that there is some $\tau \in H$ with $\tau \neq 1$ but $\tau(i) = i$ for some $i \in \{1, 2, \dots, n\}$. Since $\tau \in H \cap G_i$ and $H \cap G_i \leq G_i$, by the simplicity of G_i we must have $H \cap G_i = G_i$, that is

$$G_i \leq H.$$

By Exercise 2 of Section 1, $\sigma G_i \sigma^{-1} = G_{\sigma(i)}$, so for all i , $\sigma G_i \sigma^{-1} \leq \sigma H \sigma^{-1} = H$. Thus

$$G_j \leq H, \quad \text{for all } j \in \{1, 2, \dots, n\}.$$

Any $\lambda \in A_n$ may be written as a product of an even number, $2t$, of transpositions, so

$$\lambda = \lambda_1 \lambda_2 \cdots \lambda_t,$$

where λ_k is a product of two transpositions. Since $n > 4$ each $\lambda_k \in G_j$, for some j , hence

$$G = \langle G_1, G_2, \dots, G_n \rangle \leq H,$$

which is a contradiction. Therefore if $\tau \neq 1$ is an element of H then $\tau(i) \neq i$ for all $i \in \{1, 2, \dots, n\}$, i.e., no nonidentity element of H fixes any element of $\{1, 2, \dots, n\}$.

It follows that if τ_1, τ_2 are elements of H with

$$\tau_1(i) = \tau_2(i) \text{ for some } i, \text{ then } \tau_1 = \tau_2 \quad (4.2)$$

since then $\tau_2^{-1} \tau_1(i) = i$.

Suppose there exists a $\tau \in H$ such that the cycle decomposition of τ contains a cycle of length ≥ 3 , say

$$\tau = (a_1 a_2 a_3 \dots)(b_1 b_2 \dots) \dots$$

Let $\sigma \in G$ be an element with $\sigma(a_1) = a_1$, $\sigma(a_2) = a_2$ but $\sigma(a_3) \neq a_3$ (note that such a σ exists in A_n since $n \geq 5$). By Proposition 10

$$\tau_1 = \sigma \tau \sigma^{-1} = (a_1 a_2 \sigma(a_3) \dots)(\sigma(b_1) \sigma(b_2) \dots) \dots$$

so τ and τ_1 are distinct elements of H with $\tau(a_1) = \tau_1(a_1) = a_2$, contrary to (2). This proves that only 2-cycles can appear in the cycle decomposition of nonidentity elements of H .

Let $\tau \in H$ with $\tau \neq 1$, so that

$$\tau = (a_1 a_2)(a_3 a_4)(a_5 a_6) \dots$$

(note that $n \geq 6$ is used here). Let $\sigma = (a_1 a_2)(a_3 a_5) \in G$. Then

$$\tau_1 = \sigma \tau \sigma^{-1} = (a_1 a_2)(a_5 a_4)(a_3 a_6) \dots,$$

hence τ and τ_1 are distinct elements of H with $\tau(a_1) = \tau_1(a_1) = a_2$, again contrary to (2). This completes the proof of the simplicity of A_n .

EXERCISES

Let G be a group and let Ω be an infinite set.

1. Prove that A_n does not have a proper subgroup of index $< n$ for all $n \geq 5$.
2. Find all normal subgroups of S_n for all $n \geq 5$.
3. Prove that A_n is the only proper subgroup of index $< n$ in S_n for all $n \geq 5$.
4. Prove that A_n is generated by the set of all 3-cycles for each $n \geq 3$.
5. Prove that if there exists a chain of subgroups $G_1 \leq G_2 \leq \dots \leq G$ such that $G = \bigcup_{i=1}^{\infty} G_i$ and each G_i is simple then G is simple.
6. Let D be the subgroup of S_{Ω} consisting of permutations which move only a finite number of elements of Ω (described in Exercise 17 in Section 3) and let A be the set of all elements $\sigma \in D$ such that σ acts as an even permutation on the (finite) set of points it moves. Prove that A is an infinite simple group. [Show that every pair of elements of D lie in a finite simple subgroup of D .]
7. Under the notation of the preceding exercise prove that if $H \trianglelefteq S_{\Omega}$ and $H \neq 1$ then $A \leq H$, i.e., A is the unique (nontrivial) minimal normal subgroup of S_{Ω} .
8. Under the notation of the preceding two exercises prove that $|D| = |A| = |\Omega|$. Deduce that

$$\text{if } S_{\Omega} \cong S_{\Delta} \text{ then } |\Omega| = |\Delta|.$$

[Use the fact that D is generated by transpositions. You may assume that countable unions and finite direct products of sets of cardinality $|\Omega|$ also have cardinality $|\Omega|$.]

Direct and Semidirect Products and Abelian Groups

In this chapter we consider two of the easier methods for constructing larger groups from smaller ones, namely the notions of direct and semidirect products. This allows us to state the Fundamental Theorem on Finitely Generated Abelian Groups, which in particular completely classifies all finite abelian groups.

5.1 DIRECT PRODUCTS

We begin with the definition of the direct product of a finite and of a countable number of groups (the direct product of an arbitrary collection of groups is considered in the exercises).

Definition.

- (1) The *direct product* $G_1 \times G_2 \times \cdots \times G_n$ of the groups $G_1, G_2, \dots, G_n$ with operations $\star_1, \star_2, \dots, \star_n$, respectively, is the set of n -tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$ with operation defined componentwise:

$$(g_1, g_2, \dots, g_n) \star (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n).$$

- (2) Similarly, the *direct product* $G_1 \times G_2 \times \cdots$ of the groups $G_1, G_2, \dots$ with operations $\star_1, \star_2, \dots$, respectively, is the set of sequences $(g_1, g_2, \dots)$ where $g_i \in G_i$ with operation defined componentwise:

$$(g_1, g_2, \dots) \star (h_1, h_2, \dots) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots).$$

Although the operations may be different in each of the factors of a direct product, we shall, as usual, write all abstract groups multiplicatively, so that the operation in (1) above, for example, becomes simply

$$(g_1, g_2, \dots, g_n)(h_1, h_2, \dots, h_n) = (g_1 h_1, g_2 h_2, \dots, g_n h_n).$$

Examples

- (1) Suppose $G_i = \mathbb{R}$ (operation addition) for $i = 1, 2, \dots, n$. Then $\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$ (n -factors) is the familiar Euclidean n -space $\mathbb{R}^n$ with usual vector addition:

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n).$$

- (2) To illustrate that groups forming the direct product (and corresponding operations) may be completely general, let $G_1 = \mathbb{Z}$, let $G_2 = S_3$ and let $G_3 = GL_2(\mathbb{R})$, where the group operations are addition, composition, and matrix multiplication, respectively. Then the operation in $G_1 \times G_2 \times G_3$ is defined by

$$(n, \sigma, \begin{pmatrix} a & b \\ c & d \end{pmatrix}) (m, \tau, \begin{pmatrix} p & q \\ r & s \end{pmatrix}) = (n + m, \sigma \circ \tau, \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}).$$

Proposition 1. If $G_1, \dots, G_n$ are groups, their direct product is a group of order $|G_1| |G_2| \dots |G_n|$ (if any G_i is infinite, so is the direct product).

Proof: Let $G = G_1 \times G_2 \times \dots \times G_n$. The proof that the group axioms hold for G is straightforward since each axiom is a consequence of the fact that the same axiom holds in each factor, G_i , and the operation on G is defined componentwise. For example, the associative law is verified as follows:

Let $(a_1, a_2, \dots, a_n)$, $(b_1, b_2, \dots, b_n)$, and $(c_1, c_2, \dots, c_n) \in G$. Then

$$\begin{aligned} (a_1, a_2, \dots, a_n) [(b_1, b_2, \dots, b_n) (c_1, c_2, \dots, c_n)] \\ &= (a_1, a_2, \dots, a_n) (b_1 c_1, b_2 c_2, \dots, b_n c_n) \\ &= (a_1 (b_1 c_1), a_2 (b_2 c_2), \dots, a_n (b_n c_n)) \\ &= ((a_1 b_1) c_1, (a_2 b_2) c_2, \dots, (a_n b_n) c_n) \\ &= [(a_1, a_2, \dots, a_n) (b_1, b_2, \dots, b_n)] (c_1, c_2, \dots, c_n), \end{aligned}$$

where in the third step we have used the associative law in each component. The remaining verification that the direct product is a group is similar: the identity of G is the n -tuple $(1_1, 1_2, \dots, 1_n)$, where 1_i is the identity of G_i and the inverse of $(g_1, g_2, \dots, g_n)$ is $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$, where g_i^{-1} is the inverse of g_i in G_i .

The formula for the order of G is clear.

If the factors of the direct product are rearranged, the resulting direct product is isomorphic to the original one (cf. Exercise 7).

The next proposition shows that a direct product, $G_1 \times G_2 \times \dots \times G_n$, contains an isomorphic copy of each G_i . One can think of these specific copies as the “coordinate axes” of the direct product since, in the case of $\mathbb{R} \times \mathbb{R}$, they coincide with the x and y axes. One should be careful, however, not to think of these “coordinate axes” as the *only* copies of the groups G_i in the direct product. For example in $\mathbb{R} \times \mathbb{R}$ any line through the origin is a subgroup of $\mathbb{R} \times \mathbb{R}$ isomorphic to $\mathbb{R}$ (and $\mathbb{R} \times \mathbb{R}$ has infinitely many pairs of lines which are coordinate axes, viz. any rotation of a given coordinate system). The second part of the proposition shows that there are *projection homomorphisms* onto each of the components.

Proposition 2. Let $G_1, G_2, \dots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$ be their direct product.

- (1) For each fixed i the set of elements of G which have the identity of G_j in the j^{th} position for all $j \neq i$ and arbitrary elements of G_i in position i is a subgroup of G isomorphic to G_i :

$$G_i \cong \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\},$$

(here g_i appears in the i^{th} position). If we identify G_i with this subgroup, then $G_i \leq G$ and

$$G/G_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n.$$

- (2) For each fixed i define $\pi_i : G \rightarrow G_i$ by

$$\pi_i((g_1, g_2, \dots, g_n)) = g_i.$$

Then π_i is a surjective homomorphism with

$$\begin{aligned} \ker \pi_i &= \{(g_1, \dots, g_{i-1}, 1, g_{i+1}, \dots, g_n) \mid g_j \in G_j \text{ for all } j \neq i\} \\ &\cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n \end{aligned}$$

(here the 1 appears in position i).

- (3) Under the identifications in part (1), if $x \in G_i$ and $y \in G_j$ for some $i \neq j$, then $xy = yx$.

Proof: (1) Since the operation in G is defined componentwise, it follows easily from the subgroup criterion that $\{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$ is a subgroup of G . Furthermore, the map $g_i \mapsto (1, 1, \dots, 1, g_i, 1, \dots, 1)$ is seen to be an isomorphism of G_i with this subgroup. Identify G_i with this isomorphic copy in G .

To prove the remaining parts of (1) consider the map

$$\varphi : G \longrightarrow G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n$$

defined by

$$\varphi(g_1, g_2, \dots, g_n) = (g_1, \dots, g_{i-1}, g_{i+1}, \dots, g_n)$$

(i.e., φ erases the i^{th} component of G). The map φ is a homomorphism since

$$\begin{aligned} \varphi((g_1, \dots, g_n)(h_1, \dots, h_n)) &= \varphi((g_1 h_1, \dots, g_n h_n)) \\ &= (g_1 h_1, \dots, g_{i-1} h_{i-1}, g_{i+1} h_{i+1}, \dots, g_n h_n) \\ &= (g_1, \dots, g_{i-1}, g_{i+1}, \dots, g_n)(h_1, \dots, h_{i-1}, h_{i+1}, \dots, h_n) \\ &= \varphi((g_1, \dots, g_n))\varphi((h_1, \dots, h_n)). \end{aligned}$$

Since the entries in position j are arbitrary elements of G_j for all j , φ is surjective. Furthermore,

$$\ker \varphi = \{(g_1, \dots, g_n) \mid g_j = 1 \text{ for all } j \neq i\} = G_i.$$

This proves that G_i is a normal subgroup of G (in particular, it again proves this copy of G_i is a subgroup) and the First Isomorphism Theorem gives the final assertion of part (1).

In (2) the argument that π_i is a surjective homomorphism and the kernel is the subgroup described is very similar to that in part (1), so the details are left to the reader.

In part (3) if $x = (1, \dots, 1, g_i, 1, \dots, 1)$ and $y = (1, \dots, 1, g_j, 1, \dots, 1)$, where the indicated entries appear in positions i, j respectively, then

$$xy = (1, \dots, 1, g_i, 1, \dots, 1, g_j, 1, \dots, 1) = yx$$

(where the notation is chosen so that $i < j$). This completes the proof.

A generalization of this proposition appears as Exercise 2.

We shall continue to identify the “coordinate axis” subgroups described in part (1) of the proposition with their isomorphic copies, the G_i ’s. The i^{th} such subgroup is often called the i^{th} *component* or i^{th} *factor* of G . For instance, when we wish to calculate in $Z_n \times Z_m$ we can let x be a generator of the first factor, let y be a generator of the second factor and write the elements of $Z_n \times Z_m$ in the form $x^a y^b$. This replaces the formal ordered pairs $(x, 1)$ and $(1, y)$ with x and y (so $x^a y^b$ replaces (x^a, y^b)).

Examples

- (1) Under the notation of Proposition 2 it follows from part (3) that if $x_i \in G_i$, $1 \leq i \leq n$, then for all $k \in \mathbb{Z}$

$$(x_1 x_2 \dots x_n)^k = x_1^k x_2^k \dots x_n^k.$$

Since the order of $x_1 x_2 \dots x_n$ is the smallest positive integer k such that $x_i^k = 1$ for all i , we see that

$$|x_1 x_2 \dots x_n| = \text{l.c.m.}(|x_1|, |x_2|, \dots, |x_n|)$$

(where this order is infinite if and only if one of the x_i ’s has infinite order).

- (2) Let p be a prime and for $n \in \mathbb{Z}^+$ consider

$$E_{p^n} = Z_p \times Z_p \times \dots \times Z_p \quad (n \text{ factors}).$$

Then E_{p^n} is an abelian group of order p^n with the property that $x^p = 1$ for all $x \in E_{p^n}$. This group is the *elementary abelian* group of order p^n described in Section 4.4.

- (3) For p a prime, we show that the elementary abelian group of order p^2 has exactly $p+1$ subgroups of order p (in particular, there are more than the two obvious ones). Let $E = E_{p^2}$. Since each nonidentity element of E has order p , each of these generates a cyclic subgroup of E of order p . By Lagrange’s Theorem distinct subgroups of order p intersect trivially. Thus the $p^2 - 1$ nonidentity elements of E are partitioned into subsets of size $p - 1$ (i.e., each of these subsets consists of the nonidentity elements of some subgroup of order p). There must therefore be

$$\frac{p^2 - 1}{p - 1} = p + 1$$

subgroups of order p . When $p = 2$, E is the Klein 4-group which we have already seen has 3 subgroups of order 2 (cf. also Exercises 10 and 11).

EXERCISES

1. Show that the center of a direct product is the direct product of the centers:

$$Z(G_1 \times G_2 \times \cdots \times G_n) = Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n).$$

Deduce that a direct product of groups is abelian if and only if each of the factors is abelian.

2. Let $G_1, G_2, \dots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$. Let I be a proper, nonempty subset of $\{1, \dots, n\}$ and let $J = \{1, \dots, n\} - I$. Define G_I to be the set of elements of G that have the identity of G_j in position j for all $j \in J$.

(a) Prove that G_I is isomorphic to the direct product of the groups $G_i, i \in I$.

(b) Prove that G_I is a normal subgroup of G and $G/G_I \cong G_J$.

(c) Prove that $G \cong G_I \times G_J$.

3. Under the notation of the preceding exercise let I and K be any disjoint nonempty subsets of $\{1, 2, \dots, n\}$ and let G_I and G_K be the subgroups of G defined above. Prove that $xy = yx$ for all $x \in G_I$ and all $y \in G_K$.

4. Let A and B be finite groups and let p be a prime. Prove that any Sylow p -subgroup of $A \times B$ is of the form $P \times Q$, where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$. Prove that $n_p(A \times B) = n_p(A)n_p(B)$. Generalize both of these results to a direct product of any finite number of finite groups (so that the number of Sylow p -subgroups of a direct product is the product of the numbers of Sylow p -subgroups of the factors).

5. Exhibit a nonnormal subgroup of $Q_8 \times Z_4$ (note that every subgroup of each factor is normal).

6. Show that all subgroups of $Q_8 \times E_{2^n}$ are normal.

7. Let $G_1, G_2, \dots, G_n$ be groups and let π be a fixed element of S_n . Prove that the map

$$\varphi_\pi : G_1 \times G_2 \times \cdots \times G_n \rightarrow G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)}$$

defined by

$$\varphi_\pi(g_1, g_2, \dots, g_n) = (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$$

is an isomorphism (so that changing the order of the factors in a direct product does not change the isomorphism type).

8. Let $G_1 = G_2 = \cdots = G_n$ and let $G = G_1 \times \cdots \times G_n$. Under the notation of the preceding exercise show that $\varphi_\pi \in \text{Aut}(G)$. Show also that the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism of S_n into $\text{Aut}(G)$. (In particular, $\varphi_{\pi_1} \circ \varphi_{\pi_2} = \varphi_{\pi_1\pi_2}$. It is at this point that the π^{-1} 's in the definition of φ_π are needed. The underlying reason for this is because if e_i is the n -tuple with 1 in position i and zeros elsewhere, $1 \leq i \leq n$, then S_n acts on $\{e_1, \dots, e_n\}$ by $\pi \cdot e_i = e_{\pi(i)}$; this is a left group action. If the n -tuple $(g_1, \dots, g_n)$ is represented by $g_1e_1 + \cdots + g_ne_n$, then this left group action on $\{e_1, \dots, e_n\}$ extends to a left group action on sums by

$$\pi \cdot (g_1e_1 + g_2e_2 + \cdots + g_ne_n) = g_1e_{\pi(1)} + g_2e_{\pi(2)} + \cdots + g_ne_{\pi(n)}.$$

The coefficient of $e_{\pi(i)}$ on the right hand side is g_i , so the coefficient of e_i is $g_{\pi^{-1}(i)}$. Thus the right hand side may be rewritten as $g_{\pi^{-1}(1)}e_1 + g_{\pi^{-1}(2)}e_2 + \cdots + g_{\pi^{-1}(n)}e_n$, which is precisely the sum attached to the n -tuple $(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$. In other words, any permutation of the "position vectors" $e_1, \dots, e_n$ (which fixes their coefficients) is the same as the inverse permutation on the coefficients (fixing the e_i 's). If one uses π 's in place of π^{-1} 's in the definition of φ_π then the map $\pi \mapsto \varphi_\pi$ is not necessarily a homomorphism — it corresponds to a *right* group action.)

9. Let G_i be a field F for all i and use the preceding exercise to show that the set of $n \times n$ matrices with one 1 in each row and each column is a subgroup of $GL_n(F)$ isomorphic to S_n (these matrices are called *permutation matrices* since they simply permute the standard basis $e_1, \dots, e_n$ (as above) of the n -dimensional vector space $F \times F \times \dots \times F$).
10. Let p be a prime. Let A and B be two cyclic groups of order p with generators x and y , respectively. Set $E = A \times B$ so that E is the elementary abelian group of order p^2 : E_{p^2} . Prove that the distinct subgroups of E of order p are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \langle xy^2 \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

(note that there are $p + 1$ of them).

11. Let p be a prime and let $n \in \mathbb{Z}^+$. Find a formula for the number of subgroups of order p in the elementary abelian group E_{p^n} .
12. Let A and B be groups. Assume $Z(A)$ contains a subgroup Z_1 and $Z(B)$ contains a subgroup Z_2 with $Z_1 \cong Z_2$. Let this isomorphism be given by the map $x_i \mapsto y_i$ for all $x_i \in Z_1$. A *central product* of A and B is a quotient

$$(A \times B)/Z \quad \text{where} \quad Z = \{(x_i, y_i^{-1}) \mid x_i \in Z_1\}$$

and is denoted by $A * B$ — it is not unique since it depends on Z_1, Z_2 and the isomorphism between them. (Think of $A * B$ as the direct product of A and B “collapsed” by identifying each element $x_i \in Z_1$ with its corresponding element $y_i \in Z_2$.)

- (a) Prove that the images of A and B in the quotient group $A * B$ are isomorphic to A and B , respectively, and that these images intersect in a central subgroup isomorphic to Z_1 . Find $|A * B|$.
- (b) Let $Z_4 = \langle x \rangle$. Let $D_8 = \langle r, s \rangle$ and $Q_8 = \langle i, j \rangle$ be given by their usual generators and relations. Let $Z_4 * D_8$ be the central product of Z_4 and D_8 which identifies x^2 and r^2 (i.e., $Z_1 = \langle x^2 \rangle$, $Z_2 = \langle r^2 \rangle$ and the isomorphism is $x^2 \mapsto r^2$) and let $Z_4 * Q_8$ be the central product of Z_4 and Q_8 which identifies x^2 and -1 . Prove that $Z_4 * D_8 \cong Z_4 * Q_8$.
13. Give presentations for the groups $Z_4 * D_8$ and $Z_4 * Q_8$ constructed in the preceding exercise.
14. Let $G = A_1 \times A_2 \times \dots \times A_n$ and for each i let B_i be a normal subgroup of A_i . Prove that $B_1 \times B_2 \times \dots \times B_n \leq G$ and that
- $$(A_1 \times A_2 \times \dots \times A_n) / (B_1 \times B_2 \times \dots \times B_n) \cong (A_1/B_1) \times (A_2/B_2) \times \dots \times (A_n/B_n).$$

The following exercise describes the direct product of an arbitrary collection of groups. The terminology for the Cartesian product of an arbitrary collection of sets may be found in the Appendix.

15. Let I be any nonempty index set and let $(G_i, \star_i)$ be a group for each $i \in I$. The *direct product* of the groups $G_i, i \in I$ is the set $G = \prod_{i \in I} G_i$ (the Cartesian product of the G_i 's) with a binary operation defined as follows: if $\prod a_i$ and $\prod b_i$ are elements of G , then their product in G is given by

$$\left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} b_i \right) = \prod_{i \in I} (a_i \star_i b_i)$$

(i.e., the group operation in the direct product is defined componentwise).

- (a) Show that this binary operation is well defined and associative.
- (b) Show that the element $\prod 1_i$ satisfies the axiom for the identity of G , where 1_i is the identity of G_i for all i .

- (c) Show that the element $\prod a_i^{-1}$ is the inverse of $\prod a_i$, where the inverse of each component element a_i is taken in the group G_i .

Conclude that the direct product is a group.

(Note that if $I = \{1, 2, \dots, n\}$, this definition of the direct product is the same as the n -tuple definition in the text.)

16. State and prove the generalization of Proposition 2 to arbitrary direct products.
17. Let I be any nonempty index set and let G_i be a group for each $i \in I$. The *restricted direct product* or *direct sum* of the groups G_i is the set of elements of the direct product which are the identity in all but finitely many components, that is, the set of all elements $\prod a_i \in \prod_{i \in I} G_i$ such that $a_i = 1_i$ for all but a finite number of $i \in I$.
- Prove that the restricted direct product is a subgroup of the direct product.
 - Prove that the restricted direct product is normal in the direct product.
 - Let $I = \mathbb{Z}^+$ and let p_i be the i^{th} integer prime. Show that if $G_i = \mathbb{Z}/p_i\mathbb{Z}$ for all $i \in \mathbb{Z}^+$, then every element of the restricted direct product of the G_i 's has finite order but $\prod_{i \in \mathbb{Z}^+} G_i$ has elements of infinite order. Show that in this example the restricted direct product is the torsion subgroup of the direct product (cf. Exercise 6, Section 2.1).
18. In each of (a) to (e) give an example of a group with the specified properties:
- an infinite group in which every element has order 1 or 2
 - an infinite group in which every element has finite order but for each positive integer n there is an element of order n
 - a group with an element of infinite order and an element of order 2
 - a group G such that every finite group is isomorphic to some subgroup of G
 - a nontrivial group G such that $G \cong G \times G$.

5.2 THE FUNDAMENTAL THEOREM OF FINITELY GENERATED ABELIAN GROUPS

Definition.

- A group G is *finitely generated* if there is a finite subset A of G such that $G = \langle A \rangle$.
- For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}$ be the direct product of r copies of the group $\mathbb{Z}$, where $\mathbb{Z}^0 = 1$. The group $\mathbb{Z}^r$ is called the *free abelian group of rank r* .

Note that any finite group G is, a fortiori, finitely generated: simply take $A = G$ as a set of generators. Also, $\mathbb{Z}^r$ is finitely generated by $e_1, e_2, \dots, e_r$, where e_i is the n -tuple with 1 in position i and zeros elsewhere. We can now state the fundamental classification theorem for (finitely generated) abelian groups.

Theorem 3. (*Fundamental Theorem of Finitely Generated Abelian Groups*) Let G be a finitely generated abelian group. Then

(1)

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \dots \times \mathbb{Z}_{n_s},$$

for some integers $r, n_1, n_2, \dots, n_s$ satisfying the following conditions:

- (a) $r \geq 0$ and $n_j \geq 2$ for all j , and
 (b) $n_{i+1} \mid n_i$ for $1 \leq i \leq s-1$
 (2) the expression in (1) is unique: if $G \cong \mathbb{Z}^t \times \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_u}$, where t and $m_1, m_2, \dots, m_u$ satisfy (a) and (b) (i.e., $t \geq 0$, $m_j \geq 2$ for all j and $m_{i+1} \mid m_i$ for $1 \leq i \leq u-1$), then $t = r$, $u = s$ and $m_i = n_i$ for all i .

Proof: We shall derive this theorem in Section 12.1 as a consequence of a more general classification theorem. For finite groups we shall give an alternate proof at the end of Section 6.1.

Definition. The integer r in Theorem 3 is called the *free rank* or *Betti number* of G and the integers $n_1, n_2, \dots, n_s$ are called the *invariant factors* of G . The description of G in Theorem 3(1) is called the *invariant factor decomposition* of G .

Theorem 3 asserts that the free rank and (ordered) list of invariant factors of an abelian group are uniquely determined, so that two finitely generated abelian groups are isomorphic if and only if they have the same free rank and the same list of invariant factors. Observe that a finitely generated abelian group is a finite group if and only if its free rank is zero.

The order of a finite abelian group is just the product of its invariant factors (by Proposition 1). If G is a finite abelian group with invariant factors $n_1, n_2, \dots, n_s$, where $n_{i+1} \mid n_i$, $1 \leq i \leq s-1$, then G is said to be of *type* $(n_1, n_2, \dots, n_s)$.

Theorem 3 gives an effective way of listing *all* finite abelian groups of a given order. Namely, to find (up to isomorphism) all abelian groups of a given order n one must find all finite sequences of integers $n_1, n_2, \dots, n_s$ such that

- (1) $n_j \geq 2$ for all $j \in \{1, 2, \dots, s\}$,
 (2) $n_{i+1} \mid n_i$, $1 \leq i \leq s-1$, and
 (3) $n_1 n_2 \cdots n_s = n$.

Theorem 3 states that there is a bijection between the set of such sequences and the set of isomorphism classes of finite abelian groups of order n (where each sequence corresponds to the list of invariant factors of a finite abelian group).

Before illustrating how to find all such sequences for a specific value of n we make some general comments. First note that $n_1 \geq n_2 \geq \cdots \geq n_s$, so n_1 is the largest invariant factor. Also, by property (3) each n_i divides n . If p is any prime divisor of n then by (3) we see that p must divide n_i for some i . Then, by (2), p also divides n_j for all $j \leq i$. It follows that

every prime divisor of n must divide the first invariant factor n_1 .

In particular, if n is the product of distinct primes (all to the first power)¹ we see that $n \mid n_1$, hence $n = n_1$. This proves that if n is squarefree, there is only one possible list of invariant factors for an abelian group of order n (namely, the list $n_1 = n$):

¹Such integers are called *squarefree* since they are not divisible by any square > 1 .

Corollary 4. If n is the product of distinct primes, then up to isomorphism the only abelian group of order n is the cyclic group of order n , Z_n .

The factorization of n into prime powers is the first step in determining all possible lists of invariant factors for abelian groups of order n .

Example

Suppose $n = 180 = 2^2 \cdot 3^2 \cdot 5$. As noted above we must have $2 \cdot 3 \cdot 5 \mid n_1$, so possible values of n_1 are

$$n_1 = 2^2 \cdot 3^2 \cdot 5, \quad 2^2 \cdot 3 \cdot 5, \quad 2 \cdot 3^2 \cdot 5, \quad \text{or} \quad 2 \cdot 3 \cdot 5.$$

For each of these one must work out all possible n_2 's (subject to $n_2 \mid n_1$ and $n_1 n_2 \mid n$). For each resulting pair n_1, n_2 one must work out all possible n_3 's etc. until all lists satisfying (1) to (3) are obtained.

For instance, if $n_1 = 2 \cdot 3^2 \cdot 5$, the only number n_2 dividing n_1 with $n_1 n_2$ dividing n is $n_2 = 2$. In this case $n_1 n_2 = n$, so this list is complete: $2 \cdot 3^2 \cdot 5, 2$. The abelian group corresponding to this list is $Z_{90} \times Z_2$.

If $n_1 = 2 \cdot 3 \cdot 5$, the only candidates for n_2 are $n_2 = 2, 3$ or 6 . If $n_2 = 2$ or 3 , then since $n_3 \mid n_2$ we would necessarily have $n_3 = n_2$ (and there must be a third term in the list by property (3)). This leads to a contradiction because $n_1 n_2 n_3$ would be divisible by 2^3 or 3^3 respectively, but n is not divisible by either of these numbers. Thus the only list of invariant factors whose first term is $2 \cdot 3 \cdot 5$ is $2 \cdot 3 \cdot 5, 2 \cdot 3$. The corresponding abelian group is $Z_{30} \times Z_6$.

Similarly, all permissible lists of invariant factors and the corresponding abelian groups of order 180 are easily seen to be the following:

Invariant Factors	Abelian Groups
$2^2 \cdot 3^2 \cdot 5$	Z_{180}
$2 \cdot 3^2 \cdot 5, 2$	$Z_{90} \times Z_2$
$2^2 \cdot 3 \cdot 5, 3$	$Z_{60} \times Z_3$
$2 \cdot 3 \cdot 5, 2 \cdot 3$	$Z_{30} \times Z_6$

The process we carried out above was somewhat *ad hoc*, however it indicates that the determination of lists of invariant factors of all abelian groups of a given order n relies strongly on the factorization of n . The following theorem (which we shall see is equivalent to the Fundamental Theorem in the case of finite abelian groups) gives a more systematic and computationally much faster way of determining all finite abelian groups of a given order. More specifically, if the factorization of n is

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

it shows that all permissible lists of invariant factors for abelian groups of order n may be determined by finding permissible lists for groups of order $p_i^{\alpha_i}$ for each i . For a prime power, p^α , we shall see that the problem of determining all permissible lists is equivalent to the determination of all partitions of α (and does not depend on p).

Theorem 5. Let G be an abelian group of order $n > 1$ and let the unique factorization of n into distinct prime powers be

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

Then

- (1) $G \cong A_1 \times A_2 \times \cdots \times A_k$, where $|A_i| = p_i^{\alpha_i}$
- (2) for each $A \in \{A_1, A_2, \dots, A_k\}$ with $|A| = p^\alpha$,

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_t}}$$

with $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_t \geq 1$ and $\beta_1 + \beta_2 + \cdots + \beta_t = \alpha$ (where t and $\beta_1, \dots, \beta_t$ depend on i)

- (3) the decomposition in (1) and (2) is unique, i.e., if $G \cong B_1 \times B_2 \times \cdots \times B_m$, with $|B_i| = p_i^{\alpha_i}$ for all i , then $B_i \cong A_i$ and B_i and A_i have the same invariant factors.

Definition. The integers p^{β_j} described in the preceding theorem are called the *elementary divisors* of G . The description of G in Theorem 5(1) and 5(2) is called the *elementary divisor decomposition* of G .

The subgroups A_i described in part (1) of the theorem are the Sylow p_i -subgroups of G . Thus (1) says that G is isomorphic to the direct product of its Sylow subgroups (note that they are normal — since G is abelian — hence unique). Part 1 is often referred to as *The Primary Decomposition Theorem* for finite abelian groups.² As with Theorem 3, we shall prove this theorem later.

Note that for p a prime, $p^\beta \mid p^\gamma$ if and only if $\beta \leq \gamma$. Furthermore, $p^{\beta_1} \cdots p^{\beta_t} = p^\alpha$ if and only if $\beta_1 + \cdots + \beta_t = \alpha$. Thus the decomposition of A appearing in part (2) of Theorem 5 is the invariant factor decomposition of A with the “divisibility” conditions on the integers p^{β_j} translated into “additive” conditions on their exponents. The *elementary divisors* of G are now seen to be the *invariant factors of the Sylow p -subgroups* as p runs over all prime divisors of G .

By Theorem 5, in order to find all abelian groups of order $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ one must find for each i , $1 \leq i \leq k$, all possible lists of invariant factors for groups of order $p_i^{\alpha_i}$. The set of elementary divisors of each abelian group is then obtained by taking one set of invariant factors from each of the k lists. The abelian groups are the direct products of the cyclic groups whose orders are the elementary divisors (and distinct lists of elementary divisors give nonisomorphic groups). The advantage of this process over the one described following Theorem 2 is that it is easier to systematize how to obtain all possible lists of invariant factors, $p^{\beta_1}, p^{\beta_2}, \dots, p^{\beta_t}$, for a group of prime power order p^β . Conditions (1) to (3) for invariant factors described earlier then become

- (1) $\beta_j \geq 1$ for all $j \in \{1, 2, \dots, t\}$,
- (2) $\beta_i \geq \beta_{i+1}$ for all i , and
- (3) $\beta_1 + \beta_2 + \cdots + \beta_t = \beta$.

²Recall that for abelian groups the Sylow p -subgroups are sometimes called the p -primary components.

Hence, each list of invariant factors in this case is simply a *partition* of β (ordered in descending order). In particular, the number of nonisomorphic abelian groups of order p^β (= the number of distinct lists) equals the number of partitions of β . This number is independent of the prime p . For example the number of abelian groups of order p^5 is obtained from the list of partitions of 5:

Invariant Factors	Abelian Groups
5	Z_{p^5}
4, 1	$Z_{p^4} \times Z_p$
3, 2	$Z_{p^3} \times Z_{p^2}$
3, 1, 1	$Z_{p^3} \times Z_p \times Z_p$
2, 2, 1	$Z_{p^2} \times Z_{p^2} \times Z_p$
2, 1, 1, 1	$Z_{p^2} \times Z_p \times Z_p \times Z_p$
1, 1, 1, 1, 1	$Z_p \times Z_p \times Z_p \times Z_p \times Z_p$

Thus there are precisely 7 nonisomorphic groups of order p^5 , the first in the list being the cyclic group, Z_{p^5} , and the last in the list being the elementary abelian group, E_{p^5} .

If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ and q_i is the number of partitions of α_i , we see that the number of (distinct, nonisomorphic) abelian groups of order n equals $q_1 q_2 \cdots q_k$.

Example

If $n = 1800 = 2^3 3^2 5^2$ we list the abelian groups of this order as follows:

Order p^β	Partitions of β	Abelian Groups
2^3	3; 2, 1; 1, 1, 1	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$
3^2	2; 1, 1	$Z_9, Z_3 \times Z_3$
5^2	2; 1, 1	$Z_{25}, Z_5 \times Z_5$

We obtain the abelian groups of order 1800 by taking one abelian group from each of the three lists (right hand column above) and taking their direct product. Doing this in all possible ways gives all isomorphism types:

$Z_8 \times Z_9 \times Z_{25}$	$Z_4 \times Z_2 \times Z_3 \times Z_3 \times Z_{25}$
$Z_8 \times Z_9 \times Z_5 \times Z_5$	$Z_4 \times Z_2 \times Z_3 \times Z_3 \times Z_5 \times Z_5$
$Z_8 \times Z_3 \times Z_3 \times Z_{25}$	$Z_2 \times Z_2 \times Z_2 \times Z_9 \times Z_{25}$
$Z_8 \times Z_3 \times Z_3 \times Z_5 \times Z_5$	$Z_2 \times Z_2 \times Z_2 \times Z_9 \times Z_5 \times Z_5$
$Z_4 \times Z_2 \times Z_9 \times Z_{25}$	$Z_2 \times Z_2 \times Z_2 \times Z_3 \times Z_3 \times Z_{25}$
$Z_4 \times Z_2 \times Z_9 \times Z_5 \times Z_5$	$Z_2 \times Z_2 \times Z_2 \times Z_3 \times Z_3 \times Z_5 \times Z_5$

By the Fundamental Theorems above, this is a *complete list* of all abelian groups of order 1800 — every abelian group of this order is isomorphic to precisely one of the groups above and no two of the groups in this list are isomorphic.

We emphasize that the elementary divisors of G are not invariant factors of G (but invariant factors of *subgroups* of G). For instance, in case 1 above the elementary divisors 8, 9, 25 do not satisfy the divisibility criterion of a list of invariant factors.

Our next aim is to illustrate how to pass from a list of invariant factors of a finite abelian group to its list of elementary divisors and vice versa. We show how to determine these invariants of the group no matter how it is given as a direct product of cyclic groups. We need the following proposition.

Proposition 6. Let $m, n \in \mathbb{Z}^+$.

- (1) $Z_m \times Z_n \cong Z_{mn}$ if and only if $(m, n) = 1$.
- (2) If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ then $Z_n \cong Z_{p_1^{\alpha_1}} \times Z_{p_2^{\alpha_2}} \times \cdots \times Z_{p_k^{\alpha_k}}$.

Proof: Since (2) is an easy exercise using (1) and induction on k , we concentrate on proving (1). Let $Z_m = \langle x \rangle$, $Z_n = \langle y \rangle$ and let $l = \text{l.c.m.}(m, n)$. Note that $l = mn$ if and only if $(m, n) = 1$. Let $x^a y^b$ be a typical element of $Z_m \times Z_n$. Then (as noted in Example 1, Section 1)

$$\begin{aligned}(x^a y^b)^l &= x^{la} y^{lb} \\ &= 1^a 1^b = 1 \quad (\text{because } m \mid l \text{ and } n \mid l).\end{aligned}$$

If $(m, n) \neq 1$, every element of $Z_m \times Z_n$ has order at most l , hence has order strictly less than mn , so $Z_m \times Z_n$ cannot be isomorphic to Z_{mn} .

Conversely, if $(m, n) = 1$, then $|xy| = \text{l.c.m.}(|x|, |y|) = mn$. Thus, by order considerations, $Z_m \times Z_n = \langle xy \rangle$ is cyclic, completing the proof.

Obtaining Elementary Divisors from Invariant Factors

Suppose G is given as an abelian group of type $(n_1, n_2, \dots, n_s)$, that is

$$G \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}.$$

Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k} = n_1 n_2 \cdots n_s$. Factor each n_i as

$$n_i = p_1^{\beta_{i1}} p_2^{\beta_{i2}} \cdots p_k^{\beta_{ik}}, \quad \text{where } \beta_{ij} \geq 0.$$

By the proposition above,

$$Z_{n_i} \cong Z_{p_1^{\beta_{i1}}} \times \cdots \times Z_{p_k^{\beta_{ik}}},$$

for each i . If $\beta_{ij} = 0$, $Z_{p_j^{\beta_{ij}}} = 1$ and this factor may be deleted from the direct product without changing the isomorphism type. Then the elementary divisors of G are precisely the integers

$$p_j^{\beta_{ij}}, \quad 1 \leq j \leq k, \quad 1 \leq i \leq s \text{ such that } \beta_{ij} \neq 0.$$

For example, if $|G| = 2^3 \cdot 3^2 \cdot 5^2$ and G is of type $(30, 30, 2)$, then

$$G \cong Z_{30} \times Z_{30} \times Z_2.$$

Since $Z_{30} \cong Z_2 \times Z_3 \times Z_5$, $G \cong Z_2 \times Z_3 \times Z_5 \times Z_2 \times Z_3 \times Z_5 \times Z_2$. The elementary divisors of G are therefore 2, 3, 5, 2, 3, 5, 2, or, grouping like primes together (note that rearranging the order of the factors in a direct product does not affect the isomorphism type (Exercise 7 of Section 1)), 2, 2, 2, 3, 3, 5, 5. In particular, G is isomorphic to the last group in the list in the example above.

If for each j one collects all the factors $Z_{p_j^{a_{ij}}}$ together, the resulting direct product forms the Sylow p_j -subgroup, A_j , of G . Thus the Sylow 2-subgroup of the group in the preceding paragraph is isomorphic to $Z_2 \times Z_2 \times Z_2$ (i.e., the elementary abelian group of order 8).

Obtaining Elementary Divisors from any cyclic decomposition

The same process described above will give the elementary divisors of a finite abelian group G whenever G is given as a direct product of cyclic groups (not just when the orders of the cyclic components are the invariant factors). For example, if $G = Z_6 \times Z_{15}$, the list 6, 15 is neither that of the invariant factors (the divisibility condition fails) nor that of elementary divisors (they are not prime powers). To find the elementary divisors, factor $6 = 2 \cdot 3$ and $15 = 3 \cdot 5$. Then the prime powers 2, 3, 3, 5 are the elementary divisors and

$$G \cong Z_2 \times Z_3 \times Z_3 \times Z_5.$$

Obtaining Invariant Factors from Elementary Divisors

Suppose G is an abelian group of order n , where $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ and we are given the elementary divisors of G . The invariant factors of G are obtained by following these steps:

- (1) First group all elementary divisors which are powers of the same prime together. In this way we obtain k lists of integers (one for each p_j).
- (2) In each of these k lists arrange the integers in nonincreasing order.
- (3) Among these k lists suppose that the longest (i.e., the one with the most terms) consists of t integers. Make each of the k lists of length t by appending an appropriate number of 1's at the end of each list.
- (4) For each $i \in \{1, 2, \dots, t\}$ the i^{th} invariant factor, n_i , is obtained by taking the product of the i^{th} integer in each of the t (ordered) lists.

The point of ordering the lists in this way is to ensure that we have the divisibility condition $n_{i+1} \mid n_i$.

Suppose, for example, that the elementary divisors of G are given as 2, 3, 2, 25, 3, 2 (so $|G| = 2^3 \cdot 3^2 \cdot 5^2$). Regrouping and increasing each list to have 3 ($= t$) members gives:

$p = 2$	$p = 3$	$p = 5$
2	3	25
2	3	1
2	1	1

so the invariant factors of G are $2 \cdot 3 \cdot 25$, $2 \cdot 3 \cdot 1$, $2 \cdot 1 \cdot 1$ and

$$G \cong Z_{150} \times Z_6 \times Z_2.$$

Note that this is the penultimate group in the list classifying abelian groups of order 1800 computed above.

The invariant factor decompositions of the abelian groups of order 1800 are as follows, where the i^{th} group in this list is isomorphic to the i^{th} group computed in the

previous list:

Z_{1800}	$Z_{300} \times Z_6$
$Z_{360} \times Z_5$	$Z_{60} \times Z_{30}$
$Z_{600} \times Z_3$	$Z_{450} \times Z_2 \times Z_2$
$Z_{120} \times Z_{15}$	$Z_{90} \times Z_{10} \times Z_2$
$Z_{900} \times Z_2$	$Z_{150} \times Z_6 \times Z_2$
$Z_{180} \times Z_{10}$	$Z_{30} \times Z_{30} \times Z_2.$

Using the uniqueness statements of the Fundamental Theorems 3 and 5, we can use these processes to determine whether any two direct products of finite cyclic groups are isomorphic. For instance, if one wanted to know whether $Z_6 \times Z_{15} \cong Z_{10} \times Z_9$, first determine whether they have the same order (both are of order 90) and then (the easiest way in general) determine whether they have the same elementary divisors:

$Z_6 \times Z_{15}$ has elementary divisors 2, 3, 3, 5 and is isomorphic to $Z_2 \times Z_3 \times Z_3 \times Z_5$

$Z_{10} \times Z_9$ has elementary divisors 2, 5, 9 and is isomorphic to $Z_2 \times Z_5 \times Z_9$.

The lists of elementary divisors are different so (by Theorem 5) they are not isomorphic. Note that $Z_6 \times Z_{15}$ has no element of order 9 whereas $Z_{10} \times Z_9$ does (cf. Exercise 5).

The processes we described above (with some elaboration) form a proof (via Proposition 6) that for finite abelian groups Theorems 3 and 5 are equivalent (i.e., one implies the other). We leave the details to the reader.

One can now better understand some of the power and some of the limitations of classification theorems. On one hand, given any positive integer n one can explicitly describe all abelian groups of order n , a significant achievement. On the other hand, the amount of information necessary to determine which of the isomorphism types of groups of order n a particular group belongs to may be considerable (and is large if n is divisible by large powers of primes).

We close this section with some terminology which will be useful in later sections.

Definition.

- (1) If G is a finite abelian group of type $(n_1, n_2, \dots, n_t)$, the integer t is called the *rank* of G (the free rank of G is 0 so there will be no confusion).
- (2) If G is any group, the *exponent* of G is the smallest positive integer n such that $x^n = 1$ for all $x \in G$ (if no such integer exists the exponent of G is ∞).

EXERCISES

1. In each of parts (a) to (e) give the number of nonisomorphic abelian groups of the specified order — do not list the groups: (a) order 100, (b) order 576, (c) order 1155, (d) order 42875, (e) order 2704.
2. In each of parts (a) to (e) give the lists of invariant factors for all abelian groups of the specified order:
(a) order 270, (b) order 9801, (c) order 320, (d) order 105, (e) order 44100.
3. In each of parts (a) to (e) give the lists of elementary divisors for all abelian groups of the specified order and then match each list with the corresponding list of invariant factors

found in the preceding exercise:

(a) order 270, (b) order 9801, (c) order 320, (d) order 105, (e) order 44100.

4. In each of parts (a) to (d) determine which pairs of abelian groups listed are isomorphic (here the expression $\{a_1, a_2, \dots, a_k\}$ denotes the abelian group $Z_{a_1} \times Z_{a_2} \times \dots \times Z_{a_k}$).
 - (a) $\{4, 9\}$, $\{6, 6\}$, $\{8, 3\}$, $\{9, 4\}$, $\{6, 4\}$, $\{64\}$.
 - (b) $\{2^2, 2 \cdot 3^2\}$, $\{2^2 \cdot 3, 2 \cdot 3\}$, $\{2^3 \cdot 3^2\}$, $\{2^2 \cdot 3^2, 2\}$.
 - (c) $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\}$, $\{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\}$, $\{3 \cdot 5^2, 7^2, 3 \cdot 5 \cdot 7\}$, $\{5^2 \cdot 7, 3^2 \cdot 5 \cdot 7^2\}$.
 - (d) $\{2^2 \cdot 5 \cdot 7, 2^3 \cdot 5^3, 2 \cdot 5^2\}$, $\{2^3 \cdot 5^3 \cdot 7, 2^3 \cdot 5^3\}$, $\{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\}$, $\{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3 \cdot 7\}$.
5. Let G be a finite abelian group of type $(n_1, n_2, \dots, n_t)$. Prove that G contains an element of order m if and only if $m \mid n_1$. Deduce that G is of exponent n_1 .
6. Prove that any finite group has a finite exponent. Give an example of an infinite group with finite exponent. Does a finite group of exponent m always contain an element of order m ?
7. Let p be a prime and let $A = \langle x_1 \rangle \times \langle x_2 \rangle \times \dots \times \langle x_n \rangle$ be an abelian p -group, where $|x_i| = p^{\alpha_i} > 1$ for all i . Define the p^{th} -power map

$$\varphi : A \rightarrow A \quad \text{by} \quad \varphi : x \mapsto x^p.$$

- (a) Prove that φ is a homomorphism.
 - (b) Describe the image and kernel of φ in terms of the given generators.
 - (c) Prove both $\ker \varphi$ and $A/\text{im } \varphi$ have rank n (i.e., have the same rank as A) and prove these groups are both isomorphic to the elementary abelian group, E_{p^n} , of order p^n .
8. Let A be a finite abelian group (written multiplicatively) and let p be a prime. Let

$$A^p = \{a^p \mid a \in A\} \quad \text{and} \quad A_p = \{x \mid x^p = 1\}$$
 (so A^p and A_p are the image and kernel of the p^{th} -power map, respectively).
 - (a) Prove that $A/A^p \cong A_p$. [Show that they are both elementary abelian and they have the same order.]
 - (b) Prove that the number of subgroups of A of order p equals the number of subgroups of A of index p . [Reduce to the case where A is an elementary abelian p -group.]
9. Let $A = Z_{60} \times Z_{45} \times Z_{12} \times Z_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A .

10. Let n and k be positive integers and let A be the free abelian group of rank n (written additively). Prove that A/kA is isomorphic to the direct product of n copies of $\mathbb{Z}/k\mathbb{Z}$ (here $kA = \{ka \mid a \in A\}$). [See Exercise 14, Section 1.]
11. Let G be a nontrivial finite abelian group of rank t .
 - (a) Prove that the rank of G equals the maximum of the ranks of its Sylow subgroups.
 - (b) Prove that G can be generated by t elements but no subset with fewer than t elements generates G . [One way of doing this is by using part (a) together with Exercise 7.]
12. Let n and m be positive integers with $d = (n, m)$. Let $Z_n = \langle x \rangle$ and $Z_m = \langle y \rangle$. Let A be the central product of $\langle x \rangle$ and $\langle y \rangle$ with an element of order d identified, which has presentation $\langle x, y \mid x^n = y^m = 1, xy = yx, x^{\frac{n}{d}} = y^{\frac{m}{d}} \rangle$. Describe A as a direct product of two cyclic groups.
13. Let $A = \langle x_1 \rangle \times \dots \times \langle x_r \rangle$ be a finite abelian group with $|x_i| = n_i$ for $1 \leq i \leq r$. Find a presentation for A . Prove that if G is any group containing commuting elements $g_1, \dots, g_r$ such that $g_i^{n_i} = 1$ for $1 \leq i \leq r$, then there is a unique homomorphism from A to G which sends x_i to g_i for all i .

14. For any group G define the *dual group* of G (denoted $\widehat{G}$) to be the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$. Define a group operation in $\widehat{G}$ by pointwise multiplication of functions: if χ, ψ are homomorphisms from G into the group of roots of unity then $\chi\psi$ is the homomorphism given by $(\chi\psi)(g) = \chi(g)\psi(g)$ for all $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.
- (a) Show that this operation on $\widehat{G}$ makes $\widehat{G}$ into an abelian group. [Show that the identity is the map $g \mapsto 1$ for all $g \in G$ and the inverse of $\chi \in \widehat{G}$ is the map $g \mapsto \chi(g)^{-1}$.]
- (b) If G is a finite abelian group, prove that $\widehat{\widehat{G}} \cong G$. [Write G as $\langle x_1 \rangle \times \cdots \times \langle x_r \rangle$ and if $n_i = |x_i|$ define χ_i to be the homomorphism which sends x_i to $e^{2\pi i/n_i}$ and sends x_j to 1, for all $j \neq i$. Prove χ_i has order n_i in $\widehat{G}$ and $\widehat{G} = \langle \chi_1 \rangle \times \cdots \times \langle \chi_r \rangle$.]
- (This result is often phrased: a finite abelian group is self-dual. It implies that the lattice diagram of a finite abelian group is the same when it is turned upside down. Note however that there is no *natural* isomorphism between G and its dual (the isomorphism depends on a choice of a set of generators for G). This is frequently stated in the form: a finite abelian group is *noncanonically* isomorphic to its dual.)
15. Let $G = \langle x \rangle \times \langle y \rangle$ where $|x| = 8$ and $|y| = 4$.
- (a) Find all pairs a, b in G such that $G = \langle a \rangle \times \langle b \rangle$ (where a and b are expressed in terms of x and y).
- (b) Let $H = \langle x^2y, y^2 \rangle \cong Z_4 \times Z_2$. Prove that there are no elements a, b of G such that $G = \langle a \rangle \times \langle b \rangle$ and $H = \langle a^2 \rangle \times \langle b^2 \rangle$ (i.e., one cannot pick direct product generators for G in such a way that some powers of these are direct product generators for H).
16. Prove that no finitely generated abelian group is divisible (cf. Exercise 19, Section 2.4).

5.3 TABLE OF GROUPS OF SMALL ORDER

At this point we can give a table of the isomorphism types for most of the groups of small order.

Each of the unfamiliar non-abelian groups in the table on the following page will be constructed in Section 5 on semidirect products (which will also explain the notation used for them). For the present we give generators and relations for each of them (i.e., presentations of them).

The group $Z_3 \rtimes Z_4$ of order 12 can be described by the generators and relations:

$$\langle x, y \mid x^4 = y^3 = 1, x^{-1}yx = y^{-1} \rangle,$$

namely, it has a normal Sylow 3-subgroup $\langle y \rangle$ which is inverted by an element of order 4 (x) acting by conjugation (x^2 centralizes y).

The group $(Z_3 \times Z_3) \rtimes Z_2$ has generators and relations:

$$\langle x, y, z \mid x^2 = y^3 = z^3 = 1, yz = zy, x^{-1}yx = y^{-1}, x^{-1}zx = z^{-1} \rangle,$$

namely, it has a normal Sylow 3-subgroup isomorphic to $Z_3 \times Z_3$ ($\langle y, z \rangle$) inverted by an element of order 2 (x) acting by conjugation.

The group $Z_5 \rtimes Z_4$ of order 20 has generators and relations:

$$\langle x, y \mid x^4 = y^5 = 1, x^{-1}yx = y^{-1} \rangle,$$

namely, it has a normal Sylow 5-subgroup $\langle y \rangle$ which is inverted by an element of order 4 (x) acting by conjugation (x^2 centralizes y).

7	1	Z_7	none
8	5	$Z_8, Z_4 \times Z_2,$ $Z_2 \times Z_2 \times Z_2$	D_8, Q_8
9	2	$Z_9, Z_3 \times Z_3$	none
10	2	Z_{10}	D_{10}
11	1	Z_{11}	none
12	5	$Z_{12}, Z_6 \times Z_2$	$A_4, D_{12}, Z_3 \rtimes Z_4$
13	1	Z_{13}	none
14	2	Z_{14}	D_{14}
15	1	Z_{15}	none
16	14	$Z_{16}, Z_8 \times Z_2,$ $Z_4 \times Z_4, Z_4 \times Z_2 \times Z_2,$ $Z_2 \times Z_2 \times Z_2 \times Z_2$	not listed
17	1	Z_{17}	none
18	5	$Z_{18}, Z_6 \times Z_3$	$D_{18}, S_3 \times Z_3,$ $(Z_3 \times Z_3) \rtimes Z_2$
19	1	Z_{19}	none
20	5	$Z_{20}, Z_{10} \times Z_2$	D_{20} $Z_5 \rtimes Z_4, F_{20}$

The group F_{20} of order 20 has generators and relations:

$$\langle x, y \mid x^4 = y^5 = 1, xyx^{-1} = y^2 \rangle,$$

namely, it has a normal Sylow 5-subgroup $\langle y \rangle$ which is squared by an element of order 4 (x) acting by conjugation. One can check that this group occurs as the normalizer of Sylow 5-subgroup in S_5 , e.g.,

$$F_{20} = \langle (2\,3\,5\,4), (1\,2\,3\,4\,5) \rangle.$$

This group is called the *Frobenius group* of order 20.

EXERCISE

1. Prove that D_{16} , $Z_2 \times D_8$, $Z_2 \times Q_8$, $Z_4 * D_8$, QD_{16} and M are nonisomorphic non-abelian groups of order 16 (where $Z_4 * D_8$ is described in Exercise 12, Section 1 and QD_{16} and M are described in the exercises in Section 2.5).

5.4 RECOGNIZING DIRECT PRODUCTS

So far we have seen that direct products may be used to both construct “larger” groups from “smaller” ones and to decompose finitely generated abelian groups into cyclic factors. Even certain non-abelian groups, which may be given in some other form, may be decomposed as direct products of smaller groups. The purpose of this section is to indicate a criterion to recognize when a group is the direct product of some of its subgroups and to illustrate the criterion with some examples.

Before doing so we introduce some standard notation and elementary results on commutators which will streamline the presentation and which will be used again in Chapter 6 when we consider nilpotent groups.

Definition. Let G be a group, let $x, y \in G$ and let A, B be nonempty subsets of G .

- (1) Define $[x, y] = x^{-1}y^{-1}xy$, called the *commutator* of x and y .
- (2) Define $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$, the group generated by commutators of elements from A and from B .
- (3) Define $G' = \langle [x, y] \mid x, y \in G \rangle$, the subgroup of G generated by commutators of elements from G , called the *commutator subgroup* of G .

The commutator of x and y is 1 if and only if x and y commute, which explains the terminology. The following proposition shows how commutators measure the “difference” in G between xy and yx .

Proposition 7. Let G be a group, let $x, y \in G$ and let $H \leq G$. Then

- (1) $xy = yx[x, y]$ (in particular, $xy = yx$ if and only if $[x, y] = 1$).
- (2) $H \trianglelefteq G$ if and only if $[H, G] \leq H$.
- (3) $\sigma[x, y] = [\sigma(x), \sigma(y)]$ for any automorphism σ of G , G' char G and G/G' is abelian.
- (4) G/G' is the largest abelian quotient of G in the sense that if $H \trianglelefteq G$ and G/H is abelian, then $G' \leq H$. Conversely, if $G' \leq H$, then $H \trianglelefteq G$ and G/H is abelian.
- (5) If $\varphi : G \rightarrow A$ is any homomorphism of G into an abelian group A , then φ factors through G' i.e., $G' \leq \ker \varphi$ and the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\quad} & G/G' \\ & \searrow \varphi & \downarrow \\ & & A \end{array}$$

Proof: (1) This is immediate from the definition of $[x, y]$.

(2) By definition, $H \trianglelefteq G$ if and only if $g^{-1}hg \in H$ for all $g \in G$ and all $h \in H$. For $h \in H$, $g^{-1}hg \in H$ if and only if $h^{-1}g^{-1}hg \in H$, so that $H \trianglelefteq G$ if and only if $[h, g] \in H$ for all $h \in H$ and all $g \in G$. Thus $H \trianglelefteq G$ if and only if $[H, G] \leq H$, which is (2).

(3) Let $\sigma \in \text{Aut}(G)$ be an automorphism of G and let $x, y \in G$. Then

$$\begin{aligned}\sigma([x, y]) &= \sigma(x^{-1}y^{-1}xy) \\ &= \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) \\ &= [\sigma(x), \sigma(y)].\end{aligned}$$

Thus for every commutator $[x, y]$ of G' , $\sigma([x, y])$ is again a commutator. Since σ has a 2-sided inverse, it follows that it maps the set of commutators bijectively onto itself. Since the commutators are a generating set for G' , $\sigma(G') = G'$, that is, G' char G .

To see that G/G' is abelian, let xG' and yG' be arbitrary elements of G/G' . By definition of the group operation in G/G' and since $[x, y] \in G'$ we have

$$\begin{aligned}(xG')(yG') &= (xy)G' \\ &= (yx[x, y])G' \\ &= (yx)G' = (yG')(xG'),\end{aligned}$$

which completes the proof of (3).

(4) Suppose $H \trianglelefteq G$ and G/H is abelian. Then for all $x, y \in G$ we have $(xH)(yH) = (yH)(xH)$, so

$$\begin{aligned}1H &= (xH)^{-1}(yH)^{-1}(xH)(yH) \\ &= x^{-1}y^{-1}xyH \\ &= [x, y]H.\end{aligned}$$

Thus $[x, y] \in H$ for all $x, y \in G$, so that $G' \leq H$.

Conversely, if $G' \leq H$, then since G/G' is abelian by (3), every subgroup of G/G' is normal. In particular, $H/G' \trianglelefteq G/G'$. By the Lattice Isomorphism Theorem $H \trianglelefteq G$. By the Third Isomorphism Theorem

$$G/H \cong (G/G')/(H/G')$$

hence G/H is abelian (being isomorphic to a quotient of the abelian group G/G'). This proves (4).

(5) This is (4) phrased in terms of homomorphisms.

Passing to the quotient by the commutator subgroup of G collapses all commutators to the identity so that all elements in the quotient group commute. As (4) indicates, a strong converse to this also holds: a quotient of G by H is abelian if and only if the commutator subgroup is contained in H (i.e., if and only if G' is mapped to the identity in the quotient G/H).

We shall exhibit a group (of order 96) in the next section with the property that one of the elements of its commutator subgroup *cannot* be written as a single commutator $[x, y]$ for any x and y . Thus G' does not necessarily consist only of the set of (single) commutators (but is the group *generated* by these elements).

Examples

- (1) A group G is abelian if and only if $G' = 1$.
- (2) Sometimes it is possible to compute the commutator subgroup of a group without actually calculating commutators explicitly. For instance, if $G = D_8$, then since $Z(D_8) = \langle r^2 \rangle \leq D_8$ and $D_8/Z(D_8)$ is abelian (the Klein 4-group), the commutator subgroup D_8' is a subgroup of $Z(D_8)$. Since D_8 is not itself abelian its commutator subgroup is nontrivial. The only possibility is that $D_8' = Z(D_8)$. By a similar argument, $Q_8' = Z(Q_8) = \langle -1 \rangle$. More generally, if G is any non-abelian group of order p^3 , where p is a prime, $G' = Z(G)$ and $|G'| = p$ (Exercise 7).
- (3) Let $D_{2n} = \langle r, s \mid r^n = s^2 = 1, s^{-1}rs = r^{-1} \rangle$. Since $[r, s] = r^{-2}$, we have $\langle r^{-2} \rangle = \langle r^2 \rangle \leq D_{2n}'$. Furthermore, $\langle r^2 \rangle \trianglelefteq D_{2n}$ and the images of r and s in $D_{2n}/\langle r^2 \rangle$ generate this quotient. They are commuting elements of order ≤ 2 , so the quotient is abelian and $D_{2n}' \leq \langle r^2 \rangle$. Thus $D_{2n}' = \langle r^2 \rangle$. Finally, note that if $n (= |r|)$ is odd, $\langle r^2 \rangle = \langle r \rangle$ whereas if n is even, $\langle r^2 \rangle$ is of index 2 in $\langle r \rangle$. Hence D_{2n}' is of index 2 or 4 in D_{2n} according to whether n is odd or even, respectively.
- (4) Since conjugation by $g \in G$ is an automorphism of G , $[a^g, b^g] = [a, b]^g$ for all $a, b \in G$ by (3) of the proposition, i.e., conjugates of commutators are also commutators. For example, once we exhibit an element of one cycle type in S_n as a commutator, every element of the same cycle type is also a commutator (cf. Section 4.3). For example, every 5-cycle is a commutator in S_5 as follows: labelling the vertices of a pentagon as $1, \dots, 5$ we see that $D_{10} \leq S_5$ (a subgroup of A_5 in fact). By the preceding example an element of order 5 is a commutator in D_{10} , hence also in S_5 . Explicitly, $(1\ 4\ 2\ 5\ 3) = [(1\ 2\ 3\ 4\ 5), (2\ 5)(4\ 3)]$.

The next result actually follows from the proof of Proposition 3.13 but we isolate it explicitly for reference:

Proposition 8. Let H and K be subgroups of the group G . The number of distinct ways of writing each element of the set HK in the form hk , for some $h \in H$ and $k \in K$ is $|H \cap K|$. In particular, if $H \cap K = 1$, then each element of HK can be written uniquely as a product hk , for some $h \in H$ and $k \in K$.

Proof: Exercise.

The main result of this section is the following *recognition theorem*.

Theorem 9. Suppose G is a group with subgroups H and K such that

- (1) H and K are normal in G , and
- (2) $H \cap K = 1$.

Then $HK \cong H \times K$.

Proof: Observe that by hypothesis (1), HK is a subgroup of G (see Corollary 3.15). Let $h \in H$ and let $k \in K$. Since $H \trianglelefteq G$, $k^{-1}hk \in H$, so that $h^{-1}(k^{-1}hk) \in H$. Similarly, $(h^{-1}k^{-1}h)k \in K$. Since $H \cap K = 1$ it follows that $h^{-1}k^{-1}hk = 1$, i.e., $hk = kh$ so that every element of H commutes with every element of K .

By the preceding proposition each element of HK can be written uniquely as a product hk , with $h \in H$ and $k \in K$. Thus the map

$$\begin{aligned}\varphi : HK &\rightarrow H \times K \\ hk &\mapsto (h, k)\end{aligned}$$

is well defined. To see that φ is a homomorphism note that if $h_1, h_2 \in H$ and $k_1, k_2 \in K$, then we have seen that h_2 and k_1 commute. Thus

$$(h_1k_1)(h_2k_2) = (h_1h_2)(k_1k_2)$$

and the latter product is the unique way of writing $(h_1k_1)(h_2k_2)$ in the form hk with $h \in H$ and $k \in K$. This shows that

$$\begin{aligned}\varphi(h_1k_1h_2k_2) &= \varphi(h_1h_2k_1k_2) \\ &= (h_1h_2, k_1k_2) \\ &= (h_1, k_1)(h_2, k_2) = \varphi(h_1k_1)\varphi(h_2k_2)\end{aligned}$$

so that φ is a homomorphism. The homomorphism φ is a bijection since the representation of each element of HK as a product of the form hk is unique, which proves that φ is an isomorphism.

Definition. If G is a group and H and K are normal subgroups of G with $H \cap K = 1$, we call HK the *internal direct product* of H and K . We shall (when emphasis is called for) call $H \times K$ the *external direct product* of H and K .

The distinction between internal and external direct product is (by Theorem 9) purely notational: the elements of the internal direct product are written in the form hk , whereas those of the external direct product are written as ordered pairs (h, k) . We have in previous instances passed between these. For example, when $Z_n = \langle a \rangle$ and $Z_m = \langle b \rangle$ we wrote $x = (a, 1)$ and $y = (1, b)$ so that every element of $Z_n \times Z_m$ was written in the form $x^r y^s$.

Examples

- (1) If n is a positive odd integer, we show $D_{4n} \cong D_{2n} \times Z_2$. To see this let

$$D_{4n} = \langle r, s \mid r^{2n} = s^2 = 1, srs = r^{-1} \rangle$$

be the usual presentation of D_{4n} . Let $H = \langle s, r^2 \rangle$ and let $K = \langle r^n \rangle$. Geometrically, if D_{4n} is the group of symmetries of a regular $2n$ -gon, H is the group of symmetries of the regular n -gon inscribed in the $2n$ -gon by joining vertex $2i$ to vertex $2i + 2$, for all $i \bmod 2n$ (and if one lets $r_1 = r^2$, H has the usual presentation of the dihedral group of order $2n$ with generators r_1 and s). Note that $H \leq D_{4n}$ (it has index 2). Since $|r| = 2n$, $|r^n| = 2$. Since $srs = r^{-1}$, we have $sr^n s = r^{-n} = r^n$, that is, s centralizes r^n . Since clearly r centralizes r^n , $K \leq Z(D_{4n})$. Thus $K \leq D_{4n}$. Finally, $K \not\leq H$ since r^2 has odd order (or because r^n sends vertex i into vertex $i + n$, hence does not preserve the set of even vertices of the $2n$ -gon). Thus $H \cap K = 1$ by Lagrange. Theorem 9 now completes the proof.

- (2) Let I be a subset of $\{1, 2, \dots, n\}$ and let G be the setwise stabilizer of I in S_n , i.e.,

$$G = \{\sigma \in S_n \mid \sigma(i) \in I \text{ for all } i \in I\}.$$

Let $J = \{1, 2, \dots, n\} - I$ be the complement of I and note that G is also the setwise stabilizer of J . Let H be the *pointwise* stabilizer of I and let K be the *pointwise* stabilizer of $\{1, 2, \dots, n\} - I$, i.e.,

$$H = \{\sigma \in G \mid \sigma(i) = i \text{ for all } i \in I\}$$

$$K = \{\tau \in G \mid \tau(j) = j \text{ for all } j \in J\}.$$

It is easy to see that H and K are normal subgroups of G (in fact they are kernels of the actions of G on I and J , respectively). Since any element of $H \cap K$ fixes all of $\{1, 2, \dots, n\}$, we have $H \cap K = 1$. Finally, since every element σ of G stabilizes the sets I and J , each cycle in the cycle decomposition of σ involves only elements of I or only elements of J . Thus σ may be written as a product $\sigma_I \sigma_J$, where $\sigma_I \in H$ and $\sigma_J \in K$. This proves $G = HK$. By Theorem 9, $G \cong H \times K$. Now any permutation of J can be extended to a permutation in S_n by letting it act as the identity on I . These are precisely the permutations in H (and similarly the permutations in K are the permutations of I which are the identity on J), so

$$H \cong S_J \quad K \cong S_I \quad \text{and} \quad G \cong S_m \times S_{n-m},$$

where $m = |J|$ (and, by convention, $S_0 = 1$).

- (3) Let $\sigma \in S_n$ and let I be the subset of $\{1, 2, \dots, n\}$ fixed pointwise by σ :

$$I = \{i \in \{1, 2, \dots, n\} \mid \sigma(i) = i\}.$$

If $C = C_{S_n}(\sigma)$, then by Exercise 18 of Section 4.3, C stabilizes the set I and its complement J . By the preceding example, C is isomorphic to a subgroup of $H \times K$, where H is the subgroup of all permutations in S_n fixing I pointwise and K is the set of all permutations fixing J pointwise. Note that $\sigma \in H$. Thus each element, α , of C can be written (uniquely) as $\alpha = \alpha_I \alpha_J$, for some $\alpha_I \in H$ and $\alpha_J \in K$. Note further that if τ is any permutation of $\{1, 2, \dots, n\}$ which fixes each $j \in J$ (i.e., any element of K), then σ and τ commute (since they move no common integers). Thus C contains all such τ , i.e., C contains the subgroup K . This proves that the group C consists of all elements $\alpha_I \alpha_J \in H \times K$ such that α_J is arbitrary in K and α_I commutes with σ in H :

$$C_{S_n}(\sigma) = C_H(\sigma) \times K$$

$$\cong C_{S_I}(\sigma) \times S_I.$$

In particular, if σ is an m -cycle in S_n ,

$$C_{S_n}(\sigma) = \langle \sigma \rangle \times S_{n-m}.$$

The latter group has order $m(n-m)!$, as computed in Section 4.3.

EXERCISES

Let G be a group.

1. Prove that if $x, y \in G$ then $[y, x] = [x, y]^{-1}$. Deduce that for any subsets A and B of G , $[A, B] = [B, A]$ (recall that $[A, B]$ is the *subgroup* of G generated by the commutators $[a, b]$).
2. Prove that a subgroup H of G is normal if and only if $[G, H] \leq H$.
3. Let $a, b, c \in G$. Prove that
 - (a) $[a, bc] = [a, c](c^{-1}[a, b]c)$

- (b) $[ab, c] = (b^{-1}[a, c]b)[b, c]$.
- Find the commutator subgroups of S_4 and A_4 .
 - Prove that A_n is the commutator subgroup of S_n for all $n \geq 5$.
 - Exhibit a representative of each cycle type of A_5 as a commutator in S_5 .
 - Prove that if p is a prime and P is a non-abelian group of order p^3 then $P' = Z(P)$.
 - Assume $x, y \in G$ and both x and y commute with $[x, y]$. Prove that for all $n \in \mathbb{Z}^+$, $(xy)^n = x^n y^n [y, x]^{\frac{n(n-1)}{2}}$.
 - Prove that if p is an odd prime and P is a group of order p^3 then the p^{th} power map $x \mapsto x^p$ is a homomorphism of P into $Z(P)$. If P is not cyclic, show that the kernel of the p^{th} power map has order p^2 or p^3 . Is the squaring map a homomorphism in non-abelian groups of order 8? Where is the oddness of p needed in the above proof? [Use Exercise 8.]
 - Prove that a finite abelian group is the direct product of its Sylow subgroups.
 - Prove that if $G = HK$ where H and K are characteristic subgroups of G with $H \cap K = 1$ then $\text{Aut}(G) \cong \text{Aut}(H) \times \text{Aut}(K)$. Deduce that if G is an abelian group of finite order then $\text{Aut}(G)$ is isomorphic to the direct product of the automorphism groups of its Sylow subgroups.
 - Use Theorem 4.17 to describe the automorphism group of a finite cyclic group.
 - Prove that D_{8n} is not isomorphic to $D_{4n} \times Z_2$.
 - Let $G = \{(a_{ij}) \in GL_n(F) \mid a_{ij} = 0 \text{ if } i > j, \text{ and } a_{11} = a_{22} = \cdots = a_{nn}\}$, where F is a field, be the group of upper triangular matrices all of whose diagonal entries are equal. Prove that $G \cong D \times U$, where D is the group of all nonzero multiples of the identity matrix and U is the group of upper triangular matrices with 1's down the diagonal.
 - If A and B are normal subgroups of G such that G/A and G/B are both abelian, prove that $G/(A \cap B)$ is abelian.
 - Prove that if K is a normal subgroup of G then $K' \trianglelefteq G$.
 - If K is a normal subgroup of G and K is cyclic, prove that $G' \leq C_G(K)$. [Recall that the automorphism group of a cyclic group is abelian.]
 - Let $K_1, K_2, \dots, K_n$ be non-abelian simple groups and let $G = K_1 \times K_2 \times \cdots \times K_n$. Prove that every normal subgroup of G is of the form G_I for some subset I of $\{1, 2, \dots, n\}$ (where G_I is defined in Exercise 2 of Section 1). [If $N \trianglelefteq G$ and $x = (a_1, \dots, a_n) \in N$ with some $a_i \neq 1$, then show that there is some $g_i \in G_i$ not commuting with a_i . Show $(1, \dots, g_i, \dots, 1, x) \in K_i \cap N$ and deduce $K_i \leq N$.]
 - A group H is called *perfect* if $H' = H$ (i.e., H equals its own commutator subgroup).
 - Prove that every non-abelian simple group is perfect.
 - Prove that if H and K are perfect subgroups of a group G then $\langle H, K \rangle$ is also perfect. Extend this to show that the subgroup of G generated by any collection of perfect subgroups is perfect.
 - Prove that any conjugate of a perfect subgroup is perfect.
 - Prove that any group G has a unique maximal perfect subgroup and that this subgroup is normal.
 - Let $H(F)$ be the Heisenberg group over the field F , cf. Exercise 11 of Section 1.4. Find an explicit formula for the commutator $[X, Y]$, where $X, Y \in H(F)$, and show that the commutator subgroup of $H(F)$ equals the center of $H(F)$ (cf. Section 2.2, Exercise 14).

5.5 SEMIDIRECT PRODUCTS

In this section we study the “semidirect product” of two groups H and K , which is a generalization of the notion of the direct product of H and K obtained by relaxing the requirement that both H and K be normal. This construction will enable us (in certain circumstances) to build a “larger” group from the groups H and K in such a way that G contains subgroups isomorphic to H and K , respectively, as in the case of direct products. In this case the subgroup H will be normal in G but the subgroup K will not necessarily be normal (as it is for direct products). Thus, for instance, we shall be able to construct non-abelian groups even if H and K are abelian. This construction will allow us to enlarge considerably the set of examples of groups at our disposal. As in the preceding section, we shall then prove a recognition theorem that will enable us to decompose some familiar groups into smaller “factors,” from which we shall be able to derive some classification theorems.

By way of motivation suppose we already have a group G containing subgroups H and K such that

- (a) $H \trianglelefteq G$ (but K is not necessarily normal in G), and
- (b) $H \cap K = 1$.

It is still true that HK is a subgroup of G (Corollary 3.15) and, by Proposition 8, every element of HK can be written uniquely as a product hk , for some $h \in H$ and $k \in K$, i.e., there is a bijection between HK and the collection of ordered pairs (h, k) , given by $hk \mapsto (h, k)$ (so the group H appears as the set of elements $(h, 1)$ and K appears as the set of elements $(1, k)$). Given two elements h_1k_1 and h_2k_2 of HK , we first see how to write their product (in G) in the same form:

$$\begin{aligned}(h_1k_1)(h_2k_2) &= h_1k_1h_2(k_1^{-1}k_1)k_2 \\ &= h_1(k_1h_2k_1^{-1})k_1k_2 \\ &= h_3k_3,\end{aligned}\tag{5.1}$$

where $h_3 = h_1(k_1h_2k_1^{-1})$ and $k_3 = k_1k_2$. Note that since $H \trianglelefteq G$, $k_1h_2k_1^{-1} \in H$, so $h_3 \in H$ and $k_3 \in K$.

These calculations were predicated on the assumption that there *already existed* a group G containing subgroups H and K with $H \trianglelefteq G$ and $H \cap K = 1$. The basic idea of the semidirect product is to turn this construction around, namely start with two (abstract) groups H and K and try to *define* a group containing (an isomorphic copy of) them in such a way that (a) and (b) above hold. To do this, we write equation (1), which defines the multiplication of elements in our group, in a way that makes sense even if we do not already know there is a group containing H and K as above. The point is that k_3 in equation (1) is obtained only from multiplication in K (namely k_1k_2) and h_3 is obtained from multiplying h_1 and $k_1h_2k_1^{-1}$ in H . If we can understand where the element $k_1h_2k_1^{-1}$ arises (in terms of H and K and without reference to G), then the group HK will have been described entirely in terms of H and K . We can then use this description to *define* the group HK using equation (1) to define the multiplication.

Since H is normal in G , the group K acts on H by conjugation:

$$k \cdot h = khk^{-1} \quad \text{for } h \in H, k \in K$$

(we use the symbol $\cdot$ to emphasize the action) so that (1) can be written

$$(h_1 k_1)(h_2 k_2) = (h_1 k_1 \cdot h_2)(k_1 k_2). \quad (5.2)$$

The action of K on H by conjugation gives a homomorphism φ of K into $\text{Aut}(H)$, so (2) shows that the multiplication in HK depends only on the multiplication in H , the multiplication in K and the homomorphism φ , hence is defined intrinsically in terms of H and K .

We now use this interpretation to define a group given two groups H and K and a homomorphism φ from K to $\text{Aut}(H)$ (which will turn out to define conjugation in the resulting group).

Theorem 10. Let H and K be groups and let φ be a homomorphism from K into $\text{Aut}(H)$. Let $\cdot$ denote the (left) action of K on H determined by φ . Let G be the set of ordered pairs (h, k) with $h \in H$ and $k \in K$ and define the following multiplication on G :

$$(h_1, k_1)(h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2).$$

- (1) This multiplication makes G into a group of order $|G| = |H||K|$.
- (2) The sets $\{(h, 1) \mid h \in H\}$ and $\{(1, k) \mid k \in K\}$ are subgroups of G and the maps $h \mapsto (h, 1)$ for $h \in H$ and $k \mapsto (1, k)$ for $k \in K$ are isomorphisms of these subgroups with the groups H and K respectively:

$$H \cong \{(h, 1) \mid h \in H\} \quad \text{and} \quad K \cong \{(1, k) \mid k \in K\}.$$

Identifying H and K with their isomorphic copies in G described in (2) we have

- (3) $H \trianglelefteq G$
- (4) $H \cap K = 1$
- (5) for all $h \in H$ and $k \in K$, $khk^{-1} = k \cdot h = \varphi(k)(h)$.

Proof: It is straightforward to check that G is a group under this multiplication using the fact that $\cdot$ is an action of K on H . For example, the associative law is verified as follows:

$$\begin{aligned} ((a, x)(b, y))(c, z) &= (a x \cdot b, xy)(c, z) \\ &= (a x \cdot b (xy) \cdot c, xyz) \\ &= (a x \cdot b x \cdot (y \cdot c), xyz) \\ &= (a x \cdot (b y \cdot c), xyz) \\ &= (a, x)(b y \cdot c, yz) \\ &= (a, x)((b, y)(c, z)) \end{aligned}$$

for all $(a, x), (b, y), (c, z) \in G$. We leave as an exercise the verification that $(1, 1)$ is the identity of G and that

$$(h, k)^{-1} = (k^{-1} \cdot h^{-1}, k^{-1})$$

for each $(h, k) \in G$. The order of the group G is clearly the product of the orders of H and K , which proves (1).

Let $\tilde{H} = \{(h, 1) \mid h \in H\}$ and $\tilde{K} = \{(1, k) \mid k \in K\}$. We have

$$(a, 1)(b, 1) = (a \cdot b, 1) = (ab, 1)$$

for all $a, b \in H$ and

$$(1, x)(1, y) = (1, xy)$$

for all $x, y \in K$, which show that $\tilde{H}$ and $\tilde{K}$ are subgroups of G and that the maps in (2) are isomorphisms.

It is clear that $\tilde{H} \cap \tilde{K} = 1$, which is (4). Now,

$$\begin{aligned} (1, k)(h, 1)(1, k)^{-1} &= ((1, k)(h, 1))(1, k^{-1}) \\ &= (k \cdot h, k)(1, k^{-1}) \\ &= (k \cdot h \cdot k^{-1}, k k^{-1}) \\ &= (k \cdot h, 1) \end{aligned}$$

so that identifying $(h, 1)$ with h and $(1, k)$ with k by the isomorphisms in (2) we have $khk^{-1} = k \cdot h$, which is (5).

Finally, we have just seen that (under the identifications in (2)) $K \leq N_G(H)$. Since $G = HK$ and certainly $H \leq N_G(H)$, we have $N_G(H) = G$, i.e., $H \trianglelefteq G$, which proves (3) and completes the proof.

Definition. Let H and K be groups and let φ be a homomorphism from K into $\text{Aut}(H)$. The group described in Theorem 10 is called the *semidirect product* of H and K with respect to φ and will be denoted by $H \rtimes_{\varphi} K$ (when there is no danger of confusion we shall simply write $H \rtimes K$).

The notation is chosen to remind us that the copy of H in $H \rtimes K$ is the normal “factor” and that the construction of a semidirect product is not symmetric in H and K (unlike that of a direct product). Before giving some examples we clarify exactly when the semidirect product of H and K is their direct product (in particular, we see that direct products are a special case of semidirect products). See also Exercise 1.

Proposition 11. Let H and K be groups and let $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. Then the following are equivalent:

- (1) the identity (set) map between $H \rtimes K$ and $H \times K$ is a group homomorphism (hence an isomorphism)
- (2) φ is the trivial homomorphism from K into $\text{Aut}(H)$
- (3) $K \trianglelefteq H \rtimes K$.

Proof: (1) $\Rightarrow$ (2) By definition of the group operation in $H \rtimes K$

$$(h_1, k_1)(h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2)$$

for all $h_1, h_2 \in H$ and $k_1, k_2 \in K$. By assumption (1), $(h_1, k_1)(h_2, k_2) = (h_1 h_2, k_1 k_2)$. Equating the first factors of these ordered pairs gives $k_1 \cdot h_2 = h_2$ for all $h_2 \in H$ and all $k_1 \in K$, i.e., K acts trivially on H . This is (2).

(2) $\Rightarrow$ (3) If φ is trivial, then the action of K on H is trivial, so that the elements of H commute with those of K by Theorem 10(5). In particular, H normalizes K . Since K normalizes itself, $G = HK$ normalizes K , which is (3).

(3) $\Rightarrow$ (1) If K is normal in $H \rtimes K$ then (as in the proof of Theorem 9) for all $h \in H$ and $k \in K$, $[h, k] \in H \cap K = 1$. Thus $hk = kh$ and the action of K on H is trivial. The multiplication in the semidirect product is then the same as that in the direct product:

$$(h_1, k_1)(h_2, k_2) = (h_1 h_2, k_1 k_2)$$

for all $h_1, h_2 \in H$ and $k_1, k_2 \in K$. This gives (1) and completes the proof.

Examples

In all examples H and K are groups and φ is a homomorphism from K into $\text{Aut}(H)$ with associated action of K on H denoted by a dot. Let $G = H \rtimes K$ and as in Theorem 10 we identify H and K as subgroups of G . We shall use Propositions 4.16 and 4.17 to determine homomorphisms φ for some specific groups H . In each of the following examples the proof that φ is a homomorphism is easy (since K will often be cyclic) so the details are omitted.

- (1) Let H be any abelian group (even of infinite order) and let $K = \langle x \rangle \cong Z_2$ be the group of order 2. Define $\varphi : K \rightarrow \text{Aut}(H)$ by mapping x to the automorphism of inversion on H so that the associated action is $x \cdot h = h^{-1}$, for all $h \in H$. Then G contains the subgroup H of index 2 and

$$xhx^{-1} = h^{-1} \quad \text{for all } h \in H.$$

Of particular interest is the case when H is cyclic: if $H = Z_n$, one recognizes G as D_{2n} and if $H = \mathbb{Z}$ we denote G by D_∞ .

- (2) We can generalize the preceding example in a number of ways. One way is to let H be any abelian group and to let $K = \langle x \rangle \cong Z_{2n}$ be cyclic of order $2n$. Define φ again by mapping x to inversion, so that x^2 acts as the identity on H . In G , $xhx^{-1} = h^{-1}$ and $x^2hx^{-2} = h$ for all $h \in H$. Thus $x^2 \in Z(G)$. In particular, if $H = Z_3$ and $K = Z_4$, G is a non-abelian group of order 12 which is not isomorphic to A_4 or D_{12} (since its Sylow 2-subgroup, K , is cyclic of order 4).
- (3) Following up on the preceding example let $H = \langle h \rangle \cong Z_{2^n}$ and let $K = \langle x \rangle \cong Z_4$ with $xhx^{-1} = h^{-1}$ in G . As noted above, $x^2 \in Z(G)$. Since x inverts h (i.e., inverts H), x inverts the unique subgroup $\langle z \rangle$ of order 2 in H , where $z = h^{2^{n-1}}$. Thus $xzx^{-1} = z^{-1} = z$, so x centralizes z . It follows that $z \in Z(G)$. Thus $x^2z \in Z(G)$ hence $\langle x^2z \rangle \leq G$. Let $\bar{G} = G/\langle x^2z \rangle$. Since x^2 and z are distinct commuting elements of order 2, the order of x^2z is 2, so $|\bar{G}| = \frac{1}{2}|G| = 2^{n+1}$. By factoring out the product x^2z to form $\bar{G}$ we identify x^2 and $h^{2^{n-1}}$ in the quotient. In particular, when $n = 2$, both $\bar{x}$ and $\bar{h}$ have order 4, $\bar{x}$ inverts $\bar{h}$ and $\bar{h}^2 = \bar{x}^2$. It follows that $\bar{G} \cong Q_8$ in this case. In general, one can check that $\bar{G}$ has a unique subgroup of order 2 (namely $\langle \bar{x}^2 \rangle$) which equals the center of $\bar{G}$. The group $\bar{G}$ is called the *generalized quaternion group* of order 2^{n+1} and is denoted by $Q_{2^{n+1}}$:

$$Q_{2^{n+1}} = \langle h, x \mid h^{2^n} = x^4 = 1, x^{-1}hx = h^{-1}, h^{2^{n-1}} = x^2 \rangle.$$

- (4) Let $H = \mathbb{Q}$ (under addition) and let $K = \langle x \rangle \cong \mathbb{Z}$. Define φ by mapping x to the map “multiplication by 2” on H , so that x acts on $h \in H$ by $x \cdot h = 2h$. Note that multiplication by 2 is an automorphism of H because it has a 2-sided inverse, namely

multiplication by $\frac{1}{2}$. In the group G , $\mathbb{Z} \leq \mathbb{Q}$ and the conjugate $x\mathbb{Z}x^{-1}$ of $\mathbb{Z}$ is a proper subgroup of $\mathbb{Z}$ (namely $2\mathbb{Z}$). Thus $x \notin N_G(\mathbb{Z})$ even though $x\mathbb{Z}x^{-1} \leq \mathbb{Z}$ (note that $x^{-1}\mathbb{Z}x$ is not contained in $\mathbb{Z}$). This shows that in order to prove an element g normalizes a subgroup A in an infinite group it is not sufficient in general to show that the conjugate of A by g is just contained in A (which is sufficient for finite groups).

- (5) For H any group let $K = \text{Aut}(H)$ with φ the identity map from K to $\text{Aut}(H)$. The semidirect product $H \rtimes \text{Aut}(H)$ is called the *holomorph* of H and will be denoted by $\text{Hol}(H)$. Some holomorphs are described below; verifications of these isomorphisms are given as exercises at the end of this chapter.

(a) $\text{Hol}(\mathbb{Z}_2 \times \mathbb{Z}_2) \cong S_4$.

- (b) If $|G| = n$ and $\pi : G \rightarrow S_n$ is the left regular representation (Section 4.2), then $N_{S_n}(\pi(G)) \cong \text{Hol}(G)$. In particular, since the left regular representation of a generator of $\mathbb{Z}_n$ is an n -cycle in S_n we obtain that for any n -cycle $(1\ 2 \dots n)$:

$$N_{S_n}((1\ 2 \dots n)) \cong \text{Hol}(\mathbb{Z}_n) = \mathbb{Z}_n \rtimes \text{Aut}(\mathbb{Z}_n).$$

Note that the latter group has order $n\varphi(n)$.

- (6) Let p and q be primes with $p < q$, let $H = \mathbb{Z}_q$ and let $K = \mathbb{Z}_p$. We have already seen that if p does not divide $q - 1$ then every group of order pq is cyclic (see the example following Proposition 4.16). This is consistent with the fact that if p does not divide $q - 1$, there is no nontrivial homomorphism from $\mathbb{Z}_p$ into $\text{Aut}(\mathbb{Z}_q)$ (the latter group is cyclic of order $q - 1$ by Proposition 4.17). Assume now that $p \mid q - 1$. By Cauchy's Theorem, $\text{Aut}(\mathbb{Z}_q)$ contains a subgroup of order p (which is unique because $\text{Aut}(\mathbb{Z}_q)$ is cyclic). Thus there is a nontrivial homomorphism, φ , from K into $\text{Aut}(H)$. The associated group $G = H \rtimes K$ has order pq and K is not normal in G (Proposition 11). In particular, G is non-abelian. We shall prove shortly that G is (up to isomorphism) the unique non-abelian group of order pq . If $p = 2$, G must be isomorphic to D_{2q} .
- (7) Let p be an odd prime. We construct two nonisomorphic non-abelian groups of order p^3 (we shall later prove that any non-abelian group of order p^3 is isomorphic to one of these two).

Let $H = \mathbb{Z}_p \times \mathbb{Z}_p$ and let $K = \mathbb{Z}_p$. By Proposition 4.17, $\text{Aut}(H) \cong GL_2(\mathbb{F}_p)$ and $|GL_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p)$. Since $p \mid |\text{Aut}(H)|$, by Cauchy's Theorem H has an automorphism of order p . Thus there is a nontrivial homomorphism, φ , from K into $\text{Aut}(H)$ and so the associated group $H \rtimes K$ is a non-abelian group of order p^3 . More explicitly, if $H = \langle a \rangle \times \langle b \rangle$, and x is a generator for K then x acts on a and b by

$$x \cdot a = ab \quad \text{and} \quad x \cdot b = b$$

which defines the action of x on all of H . With respect to the $\mathbb{F}_p$ -basis a, b of the 2-dimensional vector space H the action of x (which can be considered in additive notation as a nonsingular linear transformation) has matrix

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in GL_2(\mathbb{F}_p).$$

The resulting semidirect product has the presentation

$$\langle x, a, b \mid x^p = a^p = b^p = 1, ab = ba, xax^{-1} = ab, xbx^{-1} = b \rangle$$

(in fact, this group is generated by $\{x, a\}$, and is called the *Heisenberg group* over $\mathbb{Z}/p\mathbb{Z}$, cf. Exercise 25).

Next let $H = \mathbb{Z}_{p^2}$ and $K = \mathbb{Z}_p$. Again by Proposition 4.17, $\text{Aut}(H) \cong \mathbb{Z}_{p(p-1)}$, so H admits an automorphism of order p . Thus there is a nontrivial homomorphism,

φ , from K into $\text{Aut}(H)$ and so the group $H \rtimes K$ is non-abelian and of order p^3 . More explicitly, if $H = \langle y \rangle$, and x is a generator for K then x acts on y by

$$x \cdot y = y^{1+p}.$$

The resulting semidirect product has the presentation

$$\langle x, y \mid x^p = y^{p^2} = 1, xyx^{-1} = y^{1+p} \rangle.$$

These two groups are not isomorphic (the former contains no element of order p^2 , cf. Exercise 25, and the latter clearly does, namely y).

- (8) Let $H = Q_8 \times (Z_2 \times Z_2) = \langle i, j \rangle \times (\langle a \rangle \times \langle b \rangle)$ and let $K = \langle y \rangle \cong Z_3$. The map defined by

$$i \mapsto j \quad j \mapsto k = ij \quad a \mapsto b \quad b \mapsto ab$$

is easily seen to give an automorphism of H of order 3. Let φ be the homomorphism from K to $\text{Aut}(H)$ defined by mapping y to this automorphism, and let G be the associated semidirect product, so that $y \in G$ acts by

$$y \cdot i = j \quad y \cdot j = k \quad y \cdot a = b \quad y \cdot b = ab.$$

The group $G = H \rtimes K$ is a non-abelian group of order 96 with the property that the element $i^2a \in G'$ but i^2a cannot be expressed as a single commutator $[x, y]$, for any $x, y \in G$ (checking the latter assertion is an elementary calculation).

As in the case of direct products we now prove a recognition theorem for semidirect products. This theorem will enable us to “break down” or “factor” all groups of certain orders and, as a result, classify groups of those orders. The strategy is discussed in greater detail following this theorem.

Theorem 12. Suppose G is a group with subgroups H and K such that

- (1) $H \trianglelefteq G$, and
- (2) $H \cap K = 1$.

Let $\varphi : K \rightarrow \text{Aut}(H)$ be the homomorphism defined by mapping $k \in K$ to the automorphism of left conjugation by k on H . Then $HK \cong H \rtimes K$. In particular, if $G = HK$ with H and K satisfying (1) and (2), then G is the semidirect product of H and K .

Proof: Note that since $H \trianglelefteq G$, HK is a subgroup of G . By Proposition 8 every element of HK can be written uniquely in the form hk , for some $h \in H$ and $k \in K$. Thus the map $hk \mapsto (h, k)$ is a *set* bijection from HK onto $H \rtimes K$. The fact that this map is a homomorphism is the computation at the beginning of this section which led us to the formulation of the definition of the semidirect product.

Definition. Let H be a subgroup of the group G . A subgroup K of G is called a *complement* for H in G if $G = HK$ and $H \cap K = 1$.

With this terminology, the criterion for recognizing a semidirect product is simply that there must exist a complement for some proper *normal* subgroup of G . Not every group is the semidirect product of two of its proper subgroups (for example, if the group is simple), but as we have seen, the notion of a semidirect product greatly increases our list of known groups.

Some Classifications

We now apply Theorem 12 to classify groups of order n for certain values of n . The basic idea in each of the following arguments is to

- (a) show every group of order n has proper subgroups H and K satisfying the hypothesis of Theorem 12 with $G = HK$
- (b) find all possible isomorphism types for H and K
- (c) for each pair H, K found in (b) find all possible homomorphisms $\varphi : K \rightarrow \text{Aut}(H)$
- (d) for each triple H, K, φ found in (c) form the semidirect product $H \rtimes K$ (so any group G of order n is isomorphic to one of these explicitly constructed groups) and among all these semidirect products determine which pairs are isomorphic. This results in a list of the distinct isomorphism types of groups of order n .

In order to start this process we must first find subgroups H and K (of an arbitrary group G of order n) satisfying the above conditions. In the case of “small” values of n we can often do this by Sylow’s Theorem. To show *normality* of H we use the conjugacy part of Sylow’s Theorem or other normality criteria established in Chapter 4 (e.g., Corollary 4.5). Some of this work has already been done in the examples in Section 4.5. In many of the examples that follow, $|H|$ and $|K|$ are relatively prime, so $H \cap K = 1$ holds by Lagrange’s Theorem.

Since H and K are proper subgroups of G one should think of the determination of H and K as being achieved inductively. In the examples we discuss, H and K will have sufficiently small order that we shall know all possible isomorphism types from previous results. For example, in most instances H and K will be of prime or prime squared order.

There will be relatively few possible homomorphisms $\varphi : K \rightarrow \text{Aut}(H)$ in our examples, particularly after we take into account certain symmetries (such as replacing one generator of K by another when K is cyclic).

Finally, the semidirect products which emerge from this process will, in our examples, be small in number and we shall find that, for the most part, they are (pairwise) *not* isomorphic. In general, this can be a more delicate problem, as Exercise 4 indicates.

We emphasize that this approach to “factoring” every group of some given order n as a semidirect product does not work for arbitrary n . For example, Q_8 is not a semidirect product since no proper subgroup has a complement (although we saw that it is a *quotient* of a semidirect product). Empirically, this process generally works well when the group order n is not divisible by a large power of any prime. At the other extreme, only a small percentage of the groups of order p^α for large α (p a prime) are nontrivial semidirect products.

Example: (Groups of Order pq , p and q primes with $p < q$)

Let G be any group of order pq , let $P \in \text{Syl}_p(G)$ and let $Q \in \text{Syl}_q(G)$. In Example 1 of the applications of Sylow’s Theorems we proved that $G \cong Q \rtimes P$, for some $\varphi : P \rightarrow \text{Aut}(Q)$. Since P and Q are of prime order, they are cyclic. The group $\text{Aut}(Q)$ is cyclic of order $q - 1$. If p does not divide $q - 1$, the only homomorphism from P to $\text{Aut}(Q)$ is the trivial homomorphism, hence the only semidirect product in this case is the direct product, i.e., G is cyclic.

Consider now the case when $p \mid q - 1$ and let $P = \langle y \rangle$. Since $\text{Aut}(Q)$ is cyclic it contains a unique subgroup of order p , say $\langle \gamma \rangle$, and any homomorphism $\varphi : P \rightarrow \text{Aut}(Q)$

must map y to a power of γ . There are therefore p homomorphisms $\varphi_i : P \rightarrow \text{Aut}(Q)$ given by $\varphi_i(y) = \gamma^i$, $0 \leq i \leq p-1$. Since φ_0 is the trivial homomorphism, $Q \rtimes_{\varphi_0} P \cong Q \times P$ as before. Each φ_i for $i \neq 0$ gives rise to a non-abelian group, G_i , of order pq . It is straightforward to check that these groups are all isomorphic because for each φ_i , $i > 0$, there is some generator y_i of P such that $\varphi_i(y_i) = \gamma$. Thus, up to a choice for the (arbitrary) generator of P , these semidirect products are all the same (see Exercise 6. See also Exercise 28 of Section 4.3).

Example: (Groups of Order 30)

By the examples following Sylow's Theorem every group G of order 30 contains a subgroup H of order 15. By the preceding example H is cyclic and H is normal in G (index 2). By Sylow's Theorem there is a subgroup K of G of order 2. Thus $G = HK$ and $H \cap K = 1$ so $G \cong H \rtimes K$, for some $\varphi : K \rightarrow \text{Aut}(H)$. By Proposition 4.16,

$$\text{Aut}(Z_{15}) \cong (\mathbb{Z}/15\mathbb{Z})^\times \cong Z_4 \times Z_2.$$

The latter isomorphism can be computed directly, or one can use Exercise 11 of the preceding section: writing H as $\langle a \rangle \times \langle b \rangle \cong Z_5 \times Z_3$, we have (since these two subgroups are characteristic in H)

$$\text{Aut}(H) \cong \text{Aut}(Z_5) \times \text{Aut}(Z_3).$$

In particular, $\text{Aut}(H)$ contains precisely three elements of order 2, whose actions on the group $H = \langle a \rangle \times \langle b \rangle$ are the following:

$$\left\{ \begin{array}{l} a \mapsto a \\ b \mapsto b^{-1} \end{array} \right\} \quad \left\{ \begin{array}{l} a \mapsto a^{-1} \\ b \mapsto b \end{array} \right\} \quad \left\{ \begin{array}{l} a \mapsto a^{-1} \\ b \mapsto b^{-1} \end{array} \right\}.$$

Thus there are three nontrivial homomorphisms from K into $\text{Aut}(H)$ given by sending the generator of K into one of these three elements of order 2 (as usual, the trivial homomorphism gives the direct product: $H \times K \cong Z_{30}$).

Let $K = \langle k \rangle$. If the homomorphism $\varphi_1 : K \rightarrow \text{Aut}(H)$ is defined by mapping k to the first automorphism above (so that $k \cdot a = a$ and $k \cdot b = b^{-1}$ gives the action of k on H) then $G_1 = H \rtimes_{\varphi_1} K$ is easily seen to be isomorphic to $Z_5 \times D_6$ (note that in this semidirect product k centralizes the element a of H of order 5, so the factorization as a direct product is $\langle a \rangle \times \langle b, k \rangle$).

If φ_2 is defined by mapping k to the second automorphism above, then $G_2 = H \rtimes_{\varphi_2} K$ is easily seen to be isomorphic to $Z_3 \times D_{10}$ (note that in this semidirect product k centralizes the element b of H of order 3, so the factorization as a direct product is $\langle b \rangle \times \langle a, k \rangle$).

If φ_3 is defined by mapping k to the third automorphism above then $G_3 = H \rtimes_{\varphi_3} K$ is easily seen to be isomorphic to D_{30} .

Note that these groups are all nonisomorphic since their centers have orders 30 (in the abelian case), 5 (for G_1), 3 (for G_2), and 1 (for G_3).

We emphasize that although (in hindsight) this procedure does not give rise to any groups we could not already have constructed using only direct products, the argument proves that this is the *complete* list of isomorphism types of groups of order 30.

Example: (Groups of Order 12)

Let G be a group of order 12, let $V \in \text{Syl}_2(G)$ and let $T \in \text{Syl}_3(G)$. By the discussion of groups of order 12 in Section 4.5 we know that either V or T is normal in G (for purposes of illustration we shall not invoke the full force of our results from Chapter 4, namely that either $T \trianglelefteq G$ or $G \cong A_4$). By Lagrange's Theorem $V \cap T = 1$. Thus G is a semidirect product. Note that $V \cong Z_4$ or $Z_2 \times Z_2$ and $T \cong Z_3$.

Case 1: $V \cong G$

We must determine all possible homomorphisms from T into $\text{Aut}(V)$. If $V \cong Z_4$, then $\text{Aut}(V) \cong Z_2$ and there are no nontrivial homomorphisms from T into $\text{Aut}(V)$. Thus the only group of order 12 with a normal cyclic Sylow 2-subgroup is Z_{12} .

Assume therefore that $V \cong Z_2 \times Z_2$. In this case $\text{Aut}(V) \cong S_3$ and there is a unique subgroup of $\text{Aut}(V)$ of order 3, say $\langle \gamma \rangle$. Thus if $T = \langle y \rangle$, there are three possible homomorphisms from T into $\text{Aut}(V)$:

$$\varphi_i : T \rightarrow \text{Aut}(V) \text{ defined by } \varphi_i(y) = \gamma^i, \quad i = 0, 1, 2.$$

As usual, φ_0 is the trivial homomorphism, which gives rise to the direct product $Z_2 \times Z_2 \times Z_3$. Homomorphisms φ_1 and φ_2 give rise to isomorphic semidirect products because they differ only in the choice of a generator for T (i.e., $\varphi_1(y) = \gamma$ and $\varphi_2(y') = \gamma$, where $y' = y^2$ and y' is another choice of generator for T — see also Exercise 6). The unique non-abelian group in this case is A_4 .

Case 2: $T \cong G$

We must determine all possible homomorphisms from V into $\text{Aut}(T)$. Note that $\text{Aut}(T) = \langle \lambda \rangle \cong Z_2$, where λ inverts T . If $V = \langle x \rangle \cong Z_4$, there are precisely two homomorphisms from V into $\text{Aut}(T)$: the trivial homomorphism and the homomorphism which sends x to λ . As usual, the trivial homomorphism gives rise to the direct product: $Z_3 \times Z_4 \cong Z_{12}$. The nontrivial homomorphism gives the semidirect product which was discussed in Example 2 following Proposition 11 of this section.

Finally, assume $V = \langle a \rangle \times \langle b \rangle \cong Z_2 \times Z_2$. There are precisely three nontrivial homomorphisms from V into $\text{Aut}(T)$ determined by specifying their kernels as one of the three subgroups of order 2 in V . For example, $\varphi_1(a) = \lambda$ and $\varphi_1(b) = 1$ has kernel $\langle ab \rangle$, that is, in this semidirect product both a and b act by inverting T and ab centralizes T . If φ_2 and φ_3 have kernels $\langle a \rangle$ and $\langle b \rangle$, respectively, then one easily checks that the resulting three semidirect products are all isomorphic to $S_3 \times Z_2$, where the Z_2 direct factor is the kernel of φ_i . For example,

$$T \rtimes_{\varphi_1} V = \langle a, T \rangle \times \langle ab \rangle.$$

In summary, there are precisely 5 groups of order 12, three of which are non-abelian.

Example: (Groups of Order p^3 , p an odd prime)

Let G be a group of order p^3 , p an odd prime, and assume G is not cyclic. By Exercise 9 of the previous section the map $x \mapsto x^p$ is a homomorphism from G into $Z(G)$ and the kernel of this homomorphism has order p^2 or p^3 . In the former case G must contain an element of order p^2 and in the latter case every nonidentity element of G has order p .

Case 1: G has an element of order p^2

Let x be an element of order p^2 and let $H = \langle x \rangle$. Note that since H has index p , H is normal in G by Corollary 4.5. If E is the kernel of the p^{th} power map, then in this case $E \cong Z_p \times Z_p$ and $E \cap H = \langle x^p \rangle$. Let y be any element of $E - H$ and let $K = \langle y \rangle$. By construction, $H \cap K = 1$ and so G is isomorphic to $Z_{p^2} \rtimes Z_p$, for some $\varphi : K \rightarrow \text{Aut}(H)$. If φ is the trivial homomorphism, $G \cong Z_{p^2} \times Z_p$, so we need only consider the nontrivial homomorphisms. By Proposition 4.17 $\text{Aut}(H) \cong Z_{p(p-1)}$ is cyclic and so contains a unique subgroup of order p , explicitly given by $\langle \gamma \rangle$ where

$$\gamma(x) = x^{1+p}.$$

As usual, up to choice of a generator for the cyclic group K , there is only one nontrivial homomorphism, φ , from K into $\text{Aut}(H)$, given by $\varphi(y) = \gamma$; hence up to isomorphism

there is a unique non-abelian group $H \rtimes K$ in this case. This group is described in Example 7 above.

Case 2: every nonidentity element of G has order p

In this case let H be any subgroup of G of order p^2 (see Exercise 29, Section 4.3). Necessarily $H \cong Z_p \times Z_p$. Let $K = \langle \gamma \rangle$ for any element γ of $G - H$. Since H has index p , $H \trianglelefteq G$ and since K has order p but is not contained in H , $H \cap K = 1$. Then G is isomorphic to $(Z_p \times Z_p) \rtimes Z_p$, for some $\varphi : K \rightarrow \text{Aut}(H)$. If φ is trivial, $G \cong Z_p \times Z_p \times Z_p$ (the elementary abelian group), so we may assume φ is nontrivial. By Proposition 4.17,

$$\text{Aut}(H) \cong GL_2(\mathbb{F}_p)$$

so $|\text{Aut}(H)| = (p^2 - 1)(p^2 - p)$. Note that a Sylow p -subgroup of $\text{Aut}(H)$ has order p so all subgroups of order p in $\text{Aut}(H)$ are conjugate in $\text{Aut}(H)$ by Sylow's Theorem. Explicitly, (as discussed in Example 7 above) every subgroup of order p in $\text{Aut}(H)$ is conjugate to $\langle \gamma \rangle$, where if $H = \langle a \rangle \times \langle b \rangle$, the automorphism γ is defined by

$$\gamma(a) = ab \quad \text{and} \quad \gamma(b) = b.$$

With respect to the $\mathbb{F}_p$ -basis a, b of the 2-dimensional vector space H the automorphism has matrix

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in GL_2(\mathbb{F}_p).$$

Thus (again quoting Exercise 6) there is a unique isomorphism type of semidirect product in this case.

Finally, since the two non-abelian groups have different orders for the kernels of the p^{th} power maps, they are not isomorphic. A presentation for this group is also given in Example 7 above.

EXERCISES

Let H and K be groups, let φ be a homomorphism from K into $\text{Aut}(H)$ and, as usual, identify H and K as subgroups of $G = H \rtimes_{\varphi} K$.

1. Prove that $C_K(H) = \ker \varphi$ (recall that $C_K(H) = C_G(H) \cap K$).
2. Prove that $C_H(K) = N_H(K)$.
3. In Example 1 following the proof of Proposition 11 prove that every element of $G - H$ has order 2. Prove that G is abelian if and only if $h^2 = 1$ for all $h \in H$.
4. Let $p = 2$ and check that the construction of the two non-abelian groups of order p^3 is valid in this case. Prove that *both* resulting groups are isomorphic to D_8 .
5. Let $G = \text{Hol}(Z_2 \times Z_2)$.
 - (a) Prove that $G = H \rtimes K$ where $H = Z_2 \times Z_2$ and $K \cong S_3$. Deduce that $|G| = 24$.
 - (b) Prove that G is isomorphic to S_4 . [Obtain a homomorphism from G into S_4 by letting G act on the left cosets of K . Use Exercise 1 to show this representation is faithful.]
6. Assume that K is a cyclic group, H is an arbitrary group and φ_1 and φ_2 are homomorphisms from K into $\text{Aut}(H)$ such that $\varphi_1(K)$ and $\varphi_2(K)$ are conjugate subgroups of $\text{Aut}(H)$. If K is infinite assume φ_1 and φ_2 are injective. Prove by constructing an explicit isomorphism that $H \rtimes_{\varphi_1} K \cong H \rtimes_{\varphi_2} K$ (in particular, if the subgroups $\varphi_1(K)$ and $\varphi_2(K)$ are equal in $\text{Aut}(H)$, then the resulting semidirect products are isomorphic). [Suppose $\sigma \varphi_1(K) \sigma^{-1} = \varphi_2(K)$ so that for some $a \in \mathbb{Z}$ we have $\sigma \varphi_1(k) \sigma^{-1} = \varphi_2(k)^a$ for all $k \in K$. Show that the map

$\psi : H \rtimes_{\varphi_1} K \rightarrow H \rtimes_{\varphi_2} K$ defined by $\psi((h, k)) = (\sigma(h), k^a)$ is a homomorphism. Show ψ is bijective by constructing a 2-sided inverse.]

7. This exercise describes thirteen isomorphism types of groups of order 56. (It is not too difficult to show that every group of order 56 is isomorphic to one of these.)
 - (a) Prove that there are three abelian groups of order 56.
 - (b) Prove that every group of order 56 has either a normal Sylow 2-subgroup or a normal Sylow 7-subgroup.
 - (c) Construct the following non-abelian groups of order 56 which have a normal Sylow 7-subgroup and whose Sylow 2-subgroup S is as specified:
 - one group when $S \cong Z_2 \times Z_2 \times Z_2$
 - two nonisomorphic groups when $S \cong Z_4 \times Z_2$
 - one group when $S \cong Z_8$
 - two nonisomorphic groups when $S \cong Q_8$
 - three nonisomorphic groups when $S \cong D_8$.

[For a particular S , two groups are not isomorphic if the kernels of the maps from S into $\text{Aut}(Z_7)$ are not isomorphic.]
 - (d) Let G be a group of order 56 with a nonnormal Sylow 7-subgroup. Prove that if S is the Sylow 2-subgroup of G then $S \cong Z_2 \times Z_2 \times Z_2$. [Let an element of order 7 act by conjugation on the seven nonidentity elements of S and deduce that they all have the same order.]
 - (e) Prove that there is a unique group of order 56 with a nonnormal Sylow 7-subgroup. [For existence use the fact that $|GL_3(\mathbb{F}_2)| = 168$; for uniqueness use Exercise 6.]
8. Construct a non-abelian group of order 75. Classify all groups of order 75 (there are three of them). [Use Exercise 6 to show that the non-abelian group is unique.] (The classification of groups of order pq^2 , where p and q are primes with $p < q$ and p not dividing $q - 1$, is quite similar.)
9. Show that the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 4 \end{pmatrix}$ is an element of order 5 in $GL_2(\mathbb{F}_{19})$. Use this matrix to construct a non-abelian group of order 1805 and give a presentation of this group. Classify groups of order 1805 (there are three isomorphism types). [Use Exercise 6 to prove uniqueness of the non-abelian group.] (A general method for finding elements of prime order in $GL_n(\mathbb{F}_p)$ is described in the exercises in Section 12.2; this particular matrix of order 5 in $GL_2(\mathbb{F}_{19})$ appears in Exercise 16 of that section as an illustration of the method.)
10. This exercise classifies the groups of order 147 (there are six isomorphism types).
 - (a) Prove that there are two abelian groups of order 147.
 - (b) Prove that every group of order 147 has a normal Sylow 7-subgroup.
 - (c) Prove that there is a unique non-abelian group whose Sylow 7-subgroup is cyclic.
 - (d) Let $t_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $t_2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ be elements of $GL_2(\mathbb{F}_7)$. Prove $P = \langle t_1, t_2 \rangle$ is a Sylow 3-subgroup of $GL_2(\mathbb{F}_7)$ and that $P \cong Z_3 \times Z_3$. Deduce that every subgroup of $GL_2(\mathbb{F}_7)$ of order 3 is conjugate in $GL_2(\mathbb{F}_7)$ to a subgroup of P .
 - (e) By Example 3 in Section 1 the group P has four subgroups of order 3 and these are: $P_1 = \langle t_1 \rangle$, $P_2 = \langle t_2 \rangle$, $P_3 = \langle t_1 t_2 \rangle$, and $P_4 = \langle t_1 t_2^2 \rangle$. For $i = 1, 2, 3, 4$ let $G_i = (Z_7 \times Z_7) \rtimes_{\varphi_i} Z_3$, where φ_i is an isomorphism of Z_3 with the subgroup P_i of $\text{Aut}(Z_7 \times Z_7)$. For each i describe G_i in terms of generators and relations. Deduce that $G_1 \cong G_2$.
 - (f) Prove that G_1 is not isomorphic to either G_3 or G_4 . [Show that the center of G_1 has

- order 7 whereas the centers of G_3 and G_4 are trivial.]
- (g) Prove that G_3 is not isomorphic to G_4 . [Show that every subgroup of order 7 in G_3 is normal in G_3 but that G_4 has subgroups of order 7 that are not normal.]
 - (h) Classify the groups of order 147 by showing that the six nonisomorphic groups described above (two from part (a), one from part (c) and G_1 , G_3 , and G_4) are all the groups of order 147. [Use Exercise 6 and part (d).] (The classification of groups of order pq^2 , where p and q are primes with $p < q$ and $p \nmid q - 1$, is quite similar.)
11. Classify groups of order 28 (there are four isomorphism types).
 12. Classify the groups of order 20 (there are five isomorphism types).
 13. Classify groups of order $4p$, where p is a prime greater than 3. [There are four isomorphism types when $p \equiv 3 \pmod{4}$ and five isomorphism types when $p \equiv 1 \pmod{4}$.]
 14. This exercise classifies the groups of order 60 (there are thirteen isomorphism types). Let G be a group of order 60, let P be a Sylow 5-subgroup of G and let Q be a Sylow 3-subgroup of G .
 - (a) Prove that if P is not normal in G then $G \cong A_5$. [See Section 4.5.]
 - (b) Prove that if $P \trianglelefteq G$ but Q is not normal in G then $G \cong A_4 \times Z_5$. [Show in this case that $P \leq Z(G)$, $G/P \cong A_4$, a Sylow 2-subgroup T of G is normal and $TQ \cong A_4$.]
 - (c) Prove that if both P and Q are normal in G then $G \cong Z_{15} \rtimes T$ where $T \cong Z_4$ or $Z_2 \times Z_2$. Show in this case that there are six isomorphism types when T is cyclic (one abelian) and there are five isomorphism types when T is the Klein 4-group (one abelian). [Use the same ideas as in the classifications of groups of orders 30 and 20.]
 15. Let p be an odd prime. Prove that every element of order 2 in $GL_2(\mathbb{F}_p)$ is conjugate to a diagonal matrix with ± 1 's on the diagonal. Classify the groups of order $2p^2$. [If A is a 2×2 matrix with $A^2 = I$ and v_1, v_2 is a basis for the underlying vector space, look at A acting on the vectors $w_1 = v_1 + v_2$ and $w_2 = v_1 - v_2$.]
 16. Show that there are exactly 4 distinct homomorphisms from Z_2 into $\text{Aut}(Z_8)$. Prove that the resulting semidirect products are the groups: $Z_8 \times Z_2$, D_{16} , the quasidihedral group QD_{16} and the modular group M (cf. the exercises in Section 2.5).
 17. Show that for any $n \geq 3$ there are exactly 4 distinct homomorphisms from Z_2 into $\text{Aut}(Z_{2^n})$. Prove that the resulting semidirect products give 4 nonisomorphic groups of order 2^{n+1} . [Recall Exercises 21 to 23 in Section 2.3.] (These four groups together with the cyclic group and the generalized quaternion group, $Q_{2^{n+1}}$, are all the groups of order 2^{n+1} which possess a cyclic subgroup of index 2.)
 18. Show that if H is any group then there is a group G that contains H as a normal subgroup with the property that for every automorphism σ of H there is an element $g \in G$ such that conjugation by g when restricted to H is the given automorphism σ , i.e., every automorphism of H is obtained as an inner automorphism of G restricted to H .
 19. Let H be a group of order n , let $K = \text{Aut}(H)$ and form $G = \text{Hol}(H) = H \rtimes K$ (where φ is the identity homomorphism). Let G act by left multiplication on the left cosets of K in G and let π be the associated permutation representation $\pi : G \rightarrow S_n$.
 - (a) Prove the elements of H are coset representatives for the left cosets of K in G and with this choice of coset representatives π restricted to H is the regular representation of H .
 - (b) Prove $\pi(G)$ is the normalizer in S_n of $\pi(H)$. Deduce that under the regular representation of any finite group H of order n , the normalizer in S_n of the image of H is isomorphic to $\text{Hol}(H)$. [Show $|G| = |N_{S_n}(\pi(H))|$ using Exercises 1 and 2 above.]
 - (c) Deduce that the normalizer of the group generated by an n -cycle in S_n is isomorphic to $\text{Hol}(Z_n)$ and has order $n\varphi(n)$.